An Introduction to Use-Case Modeling

- The process of modeling a system's functions in terms of
  - business events
  - who initiated the events
  - how the system responds to those events
- An approach that facilitates ***user-centered development***

User-Centered Development and Use-Case Modeling

User-centered development – a process of systems development based on understanding the ***needs of the stakeholders*** and the ***reasons*** why the system should be developed.

Use-case modeling – the process of modeling a system's functions in terms of business events, who initiated the events, and how the system responds to those events.

- Use-case modeling has roots in object-oriented modeling.
- Popular in non-object development environments because of its usefulness in communicating with users.

An Introduction to Use-Case Modeling

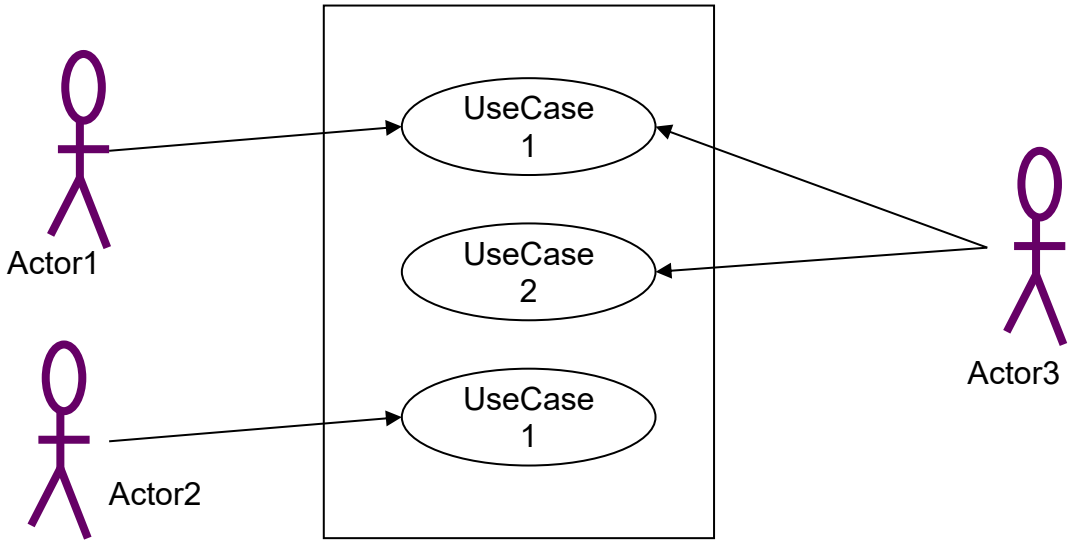
- Originally conceived by Dr. Ivar Jacobson in 1986.
- Proved to be a valuable aid in meeting the challenges of determining what a system is required to do from a user and stakeholder perspective.
- A best practice for *defining*, *documenting* and *understanding* of an information system's **functional requirements**.

An Introduction to Use-Case Modeling

- Benefits
  - Facilitates and encourages user involvement
  - Provides a tool for capturing functional requirements
  - Provides an aid in estimating project scope, effort and schedule
  - Provides a tool for *requirements traceability*
  - Provides a framework for driving the system development project

Use-case diagram

- Depicts the interactions between the **system** and **external systems / users.**
- Graphically describes ***who will use the system*** *and* ***in what ways the user expects to interact with the system***



System concepts for Use-Case Modeling

- **Use Cases**

  Needs three things

  Title – What is the goal? Short phrase with an active verb.

  Actor – Who desires it?

  Scenario – How is it accomplished?

- **Scenario –** Purchase Items

  - Customer Reviews items in the Shopping Cart
  - Customer provides payment and shopping information
  - System validate payment Information and respond with confirmation of order
  - System will send confirmation of order details to customer in an email.

User Story

Simpler and shorter way of describing a scenario from a user perspective. Popular with Agile Methods

Format:

As a <type of user>

I want <goal>

So that <Reason>

Eg. As a Bank Customer

I want to change my PIN online

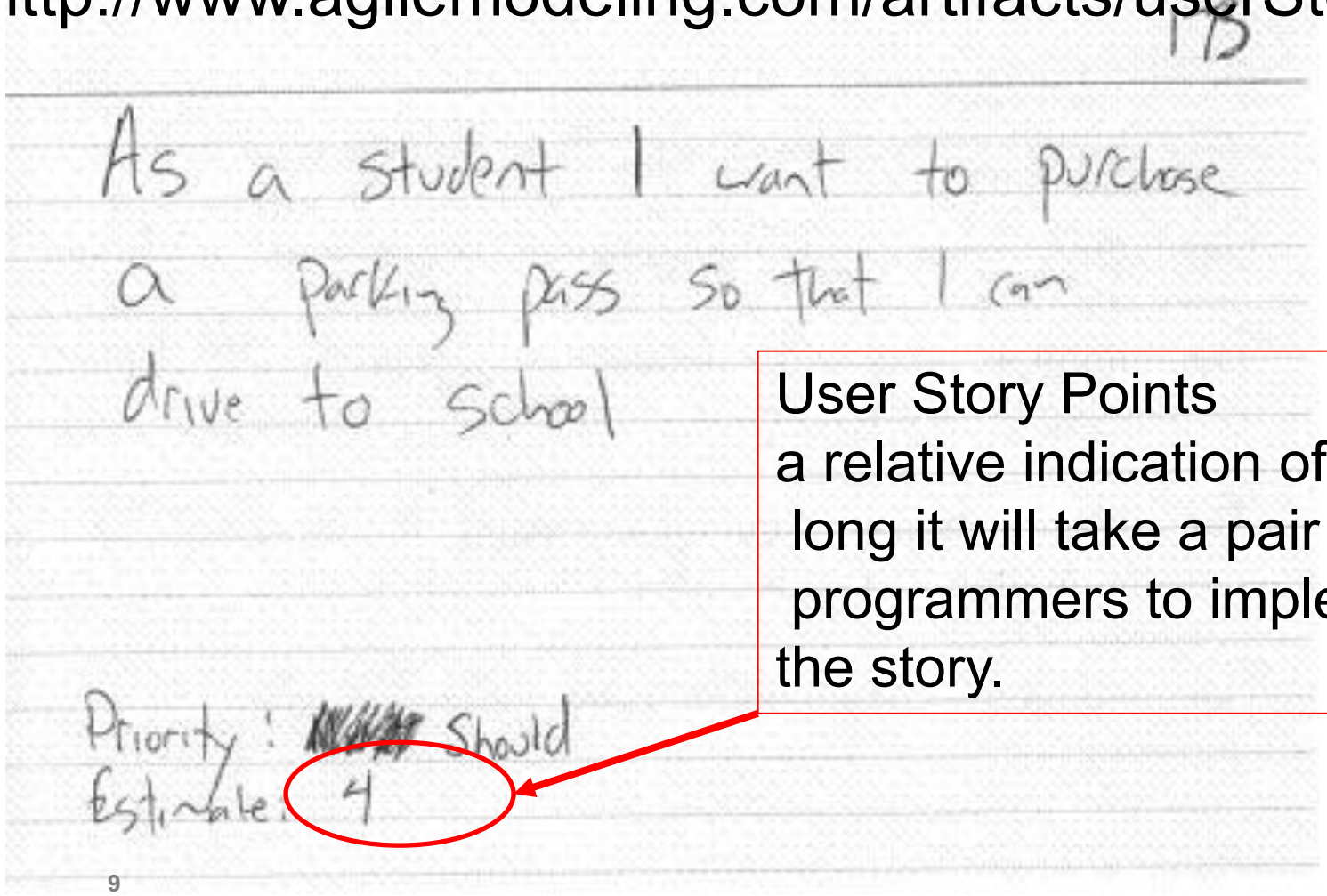
So that I do not want to go the Bank

Expresses something that
a user of the system
wants to be able to do.

A team at Connextra developed the traditional user-story template in 2001

User Story Card (Front)

http://www.agilemodeling.com/artifacts/userStory.htm



User Story Points
a relative indication of how long it will take a pair of programmers to implement the story.

User Story Card (Back)

Front of Card

1B

As a student I want to purchase a parking pass so that I can drive to school

Priority: Should
Estimate: 4

Back of Card

Confirmations:

~~The student must pay the correct amt~~

One pass for one month is issued at a time

The student will not recieve a pass if the payment isn't sufficient

The person buying the pass must be a currently enrolled student.

The student may only buy one pass per month.

Use Story

As a (late riser), I want (to get my breakfast quickly)

in order (to catch the train)

- **Can break this into number of more detailed stories about real requirements**

As a (late riser), I want (a four-slice toaster)

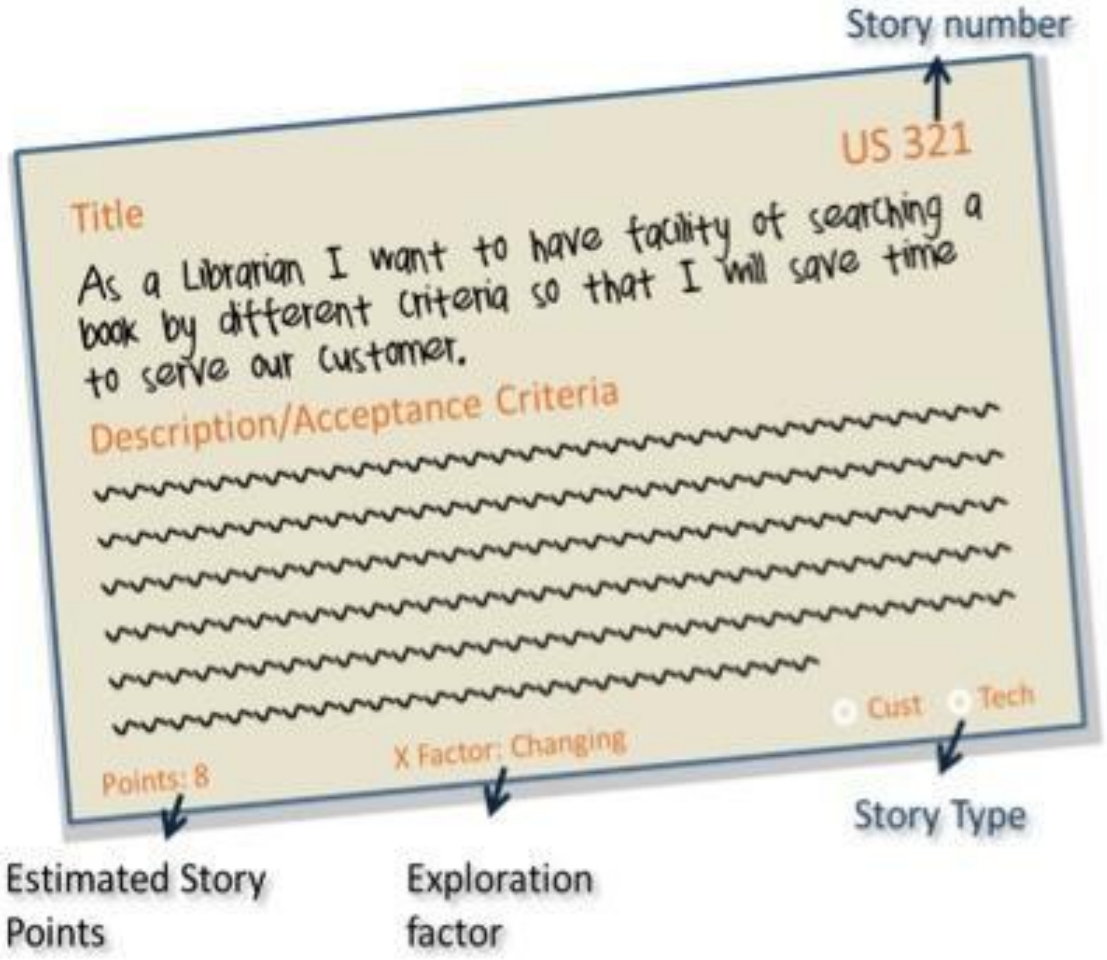
in order (to make breakfast quicker)

And

I want (a folding bike) in order (to get to the station quicker)

What are Epics ,Themes?

User Stories used in Agile Development



Electronic User Story may contain more information such as
-Responsible person
-Sprint US is part of
-Attached file or picture etc.

Also acceptance criteria might be written on the back side of the US. Exploration Factor describes how uncertain requirement is, its values can be completed, incomplete, dynamic, stable etc.

User Stories

- **User stories are about needs**. It's something that the user needs to do in his day-to-day job.
- **User stories are easy for users to read.** When you write a user story, write something that anyone can understand, in the language of the users.
- **A User story is independent of the other user stories**

USER STORIES AND USE CASES

USER STORIES	USE CASES
short - one index card	long - a document
one goal, no details	multiple goals and details
informal	casual to (very) formal
"placeholder for conversation"	"record of conversation"

System concepts for Use-Case Modeling

Use-case narrative

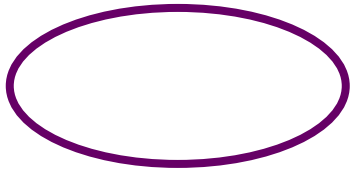
- A textual description of the business event and how the user will interact with the system to accomplish the task.
- Use active voice.

  eg. System validates Member-ID

System concepts for Use-Case Modeling

- Use Cases
  - A behaviorally related sequence of steps both automated and manual for the purpose of completing a single business task.
  - Describe system functions from the perspective of external users in a manner they understand.

System concepts for Use-Case Modeling



What is a Use Case?

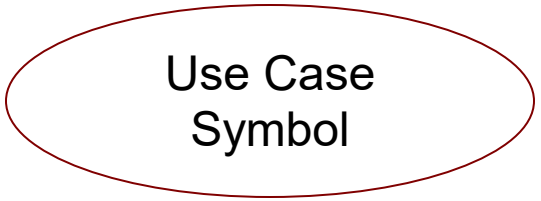
📖 It is a typical interaction between a user and a computer system.

📖 They are the primary elements in software development.

📖 They represent the functionality provided by the system. i.e. what capabilities will be provided to an actor by the system.

System concepts for Use-Case Modeling

- Use Cases
  - Represented graphically by a horizontal ellipse with the name of the use case appearing above, below, or inside the ellipse.



System concepts for Use-Case Modeling

📖 **The collection of Use Cases for a system constitute all the defined ways the system may be used.**

How to capture Use Cases?

By talking to typical users and discussing things that they might want to do with the system.

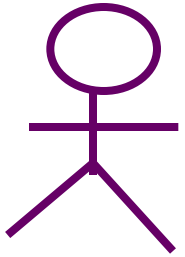
eg. Borrowing of books , Returning of books etc. (Library System)

System concepts for Use-Case Modeling

- **Actors**
  - Anyone or anything that needs to interact with the system to exchange information.
  - Represented graphically as a stick figure labeled with the name of the role the actor plays.

System concepts for Use-Case Modeling

- **Actors**



  - An Actor may
    - ➢ Only input information to the system.
    - ➢ Only receive information from the system.
    - ➢ Input and receive information to and from the system.
  - Typically they are found in the problem statement and by conversations with customers and domain experts.

    eg. *Librarian* in a Library System

    Bank

Bank

Actors

There are primarily four types of Actors

- Primary Business Actors
- Primary System Actor
- External Server Actor
- External Receiver Actor

Types of Actors

- **Primary Business Actors** – Benefits from the execution of use cases by receiving some thing measurable.

  eg. Employee receiving a pay cheque.

- **Primary System Actors** - Directly Interfaces with the system to trigger an event.

  eg. Grocery Clerk – Scan customer buying Items .

Types of Actors Cont...

- **External Server Actor**

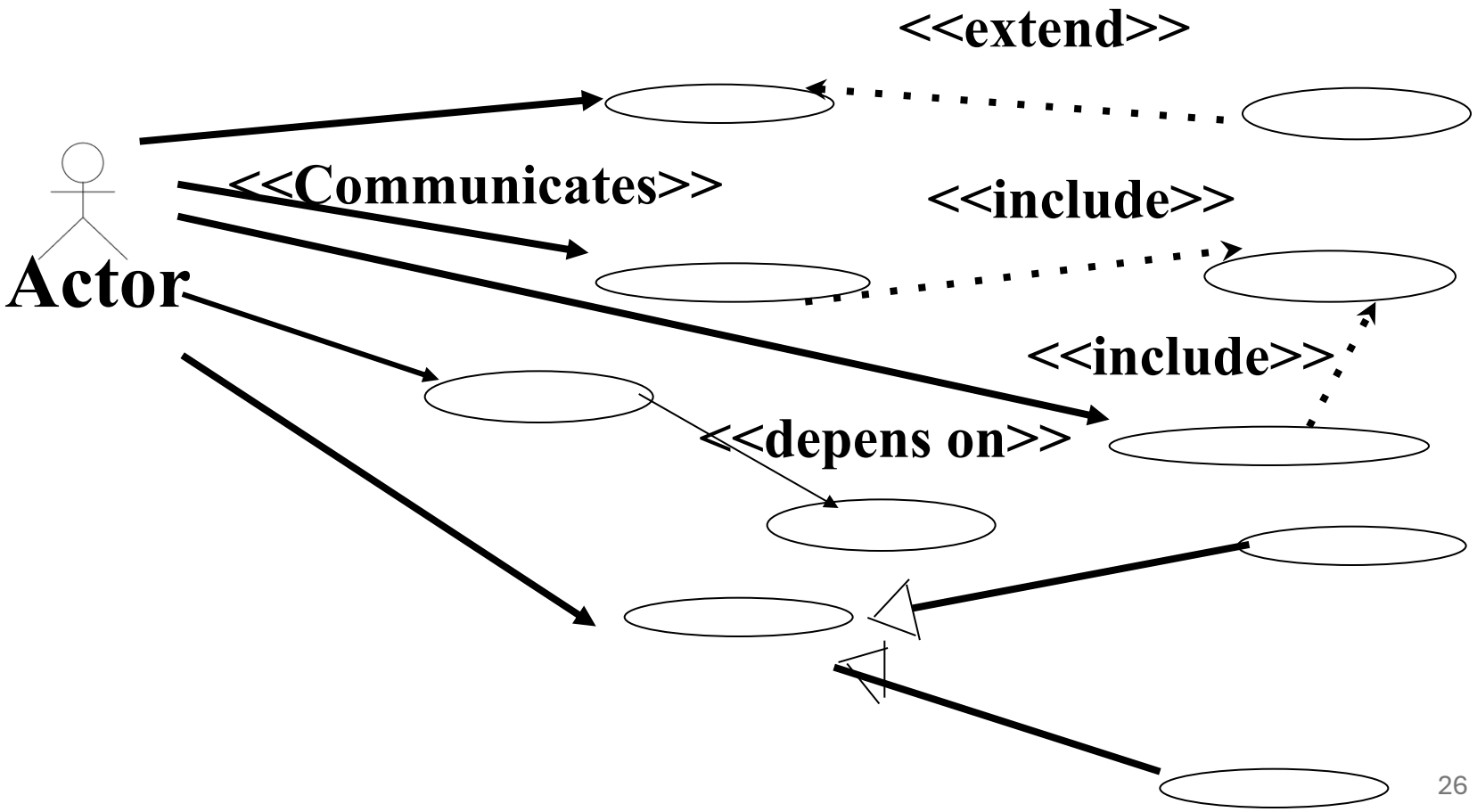
  eg. Credit bureau authorizing the charging by a credit card.

- **External Receiver Actor**

  eg. Warehouse receiving a package order to prepare a shipment.

Use Case Diagram

📖 Introduced by Jacobson (1994) to visualize use cases.



Use Case Diagram

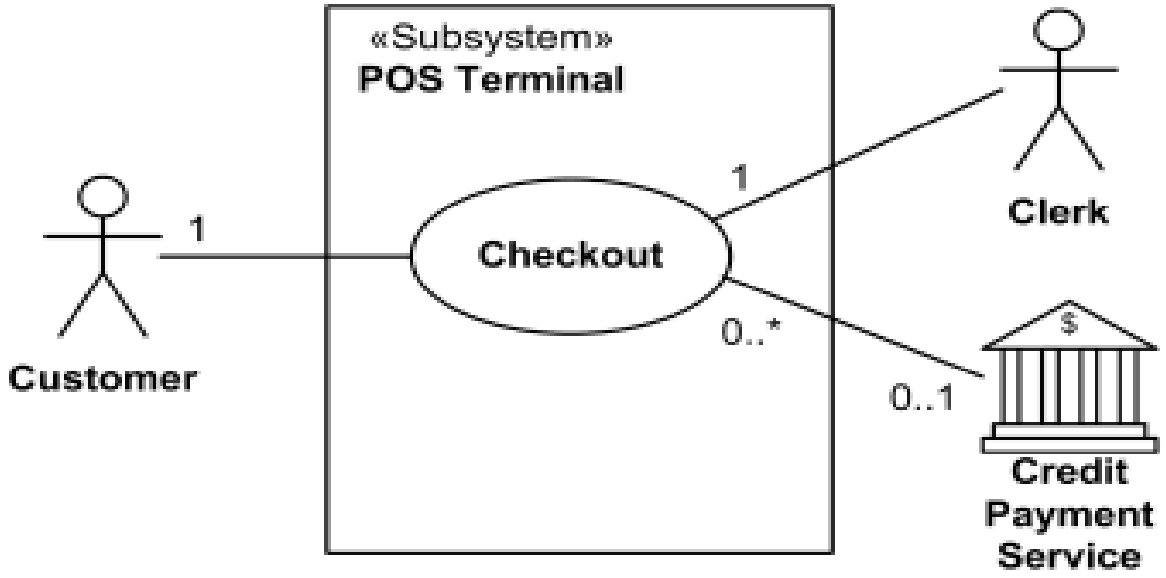
- Relationships
  - Depicted as a line between two symbols on the use-case diagram
  - Meaning of the relationship differs depending on how the lines are drawn and type of symbols they connect.

Use Case Diagram

- Relationships
  - Associations (also called <<Communicates>>)
    - A relationship between an actor and a use case in which an interaction occurs between them
    - Modeled as a solid line connecting the actor and the use case
    - May be bidirectional or unidirectional
    - UML 2.5 allows multiplicity

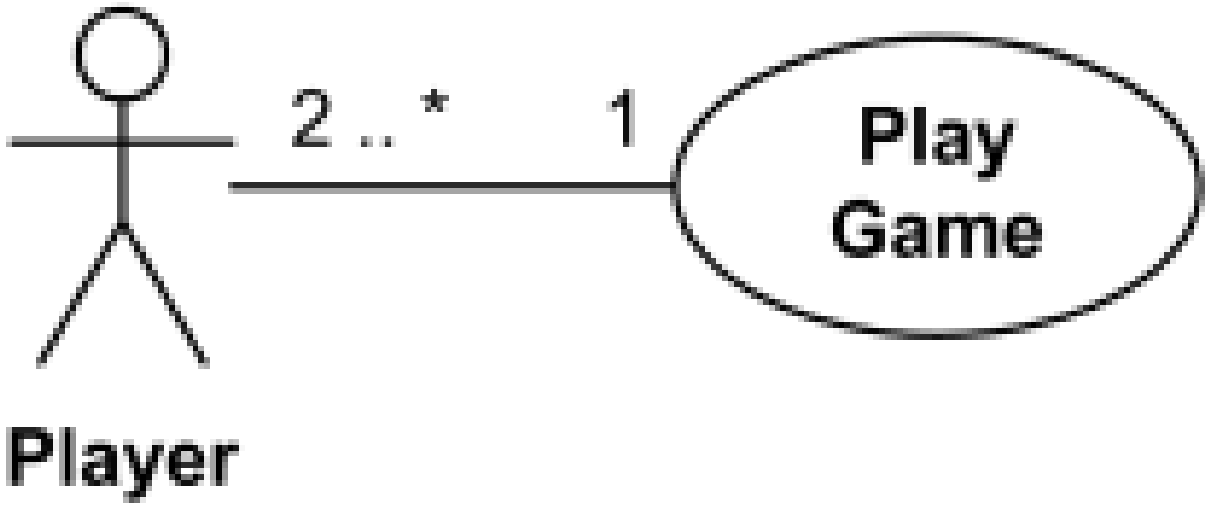
Multiplicity of an Actor/Use Case (UML 2.5)

- Required actor may be explicitly denoted using multiplicity 1 or greater.
- UML 2.5 also allows actor to be optional.
- Multiplicity 0..1 of actor means that the actor may or may not participate in any of their associated use cases.



Multiplicity of an Actor / Use Case (UML 2.5)

- When a use case has an association to an actor with a multiplicity that is greater than one at the actor end, it means that more than one actor instance is involved in the use case.

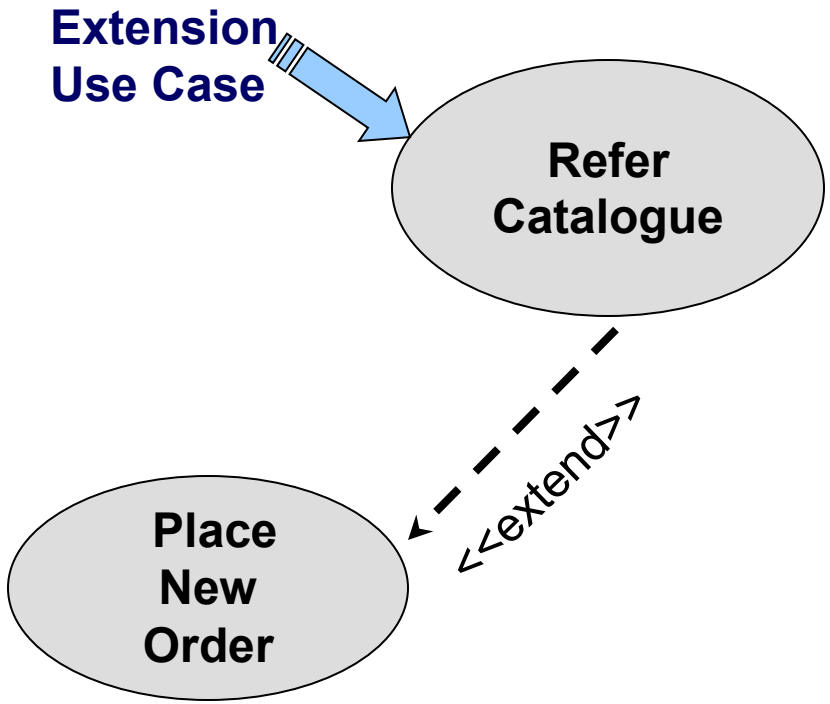


Use Case Diagram

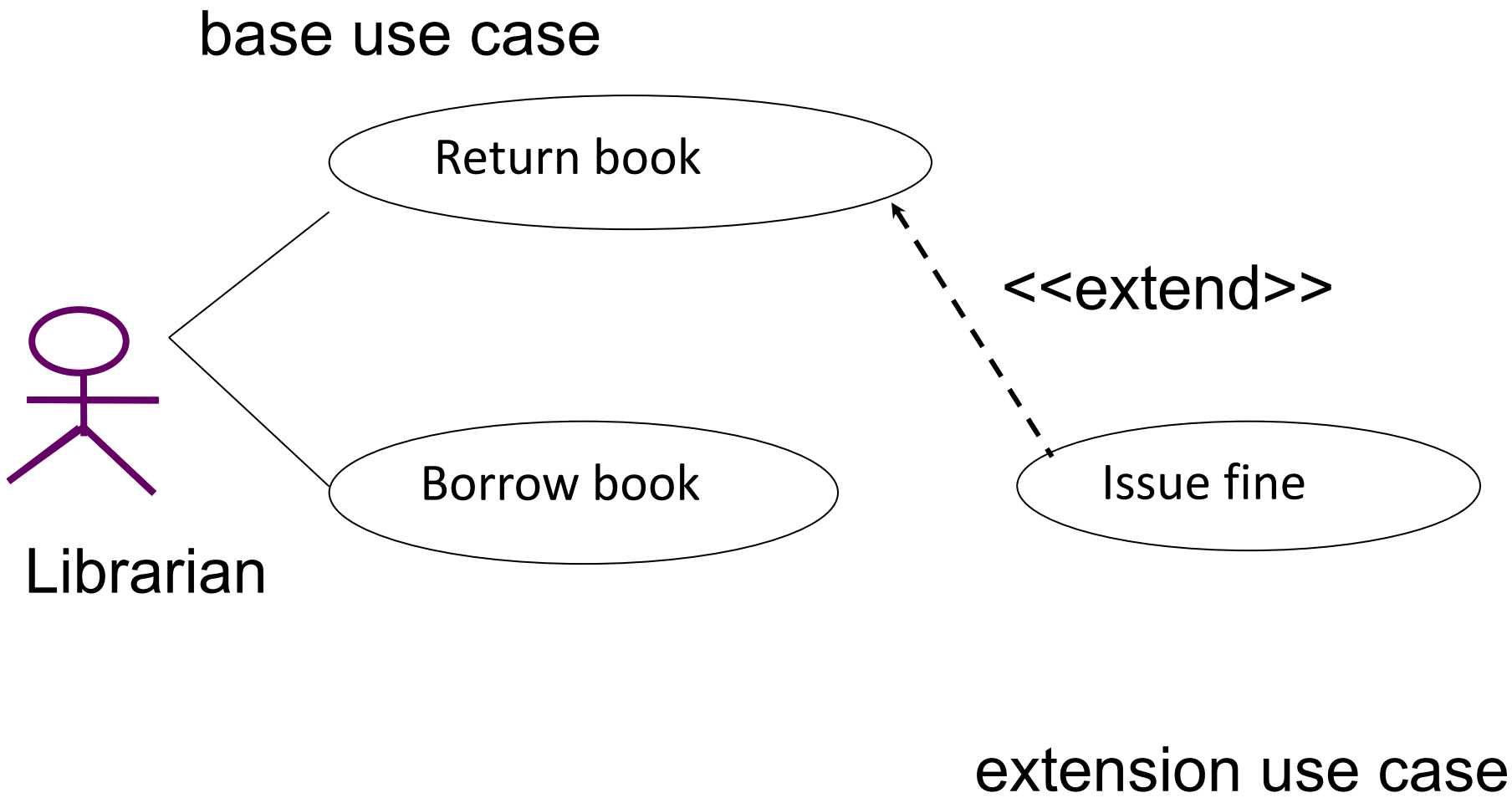
- Relationships
  - Extend **<<extend>>**
    - <<extend>> provides a way to insert new behaviour into an existing use case.
    - The extension use case extends the functionality of the original use case.
    - Shows optional behavior of a Use Case
    - Depicted as an arrow headed line (either solid/dashed)

Use Case Diagram

- Relationships
  - Extend



Use Case Diagram



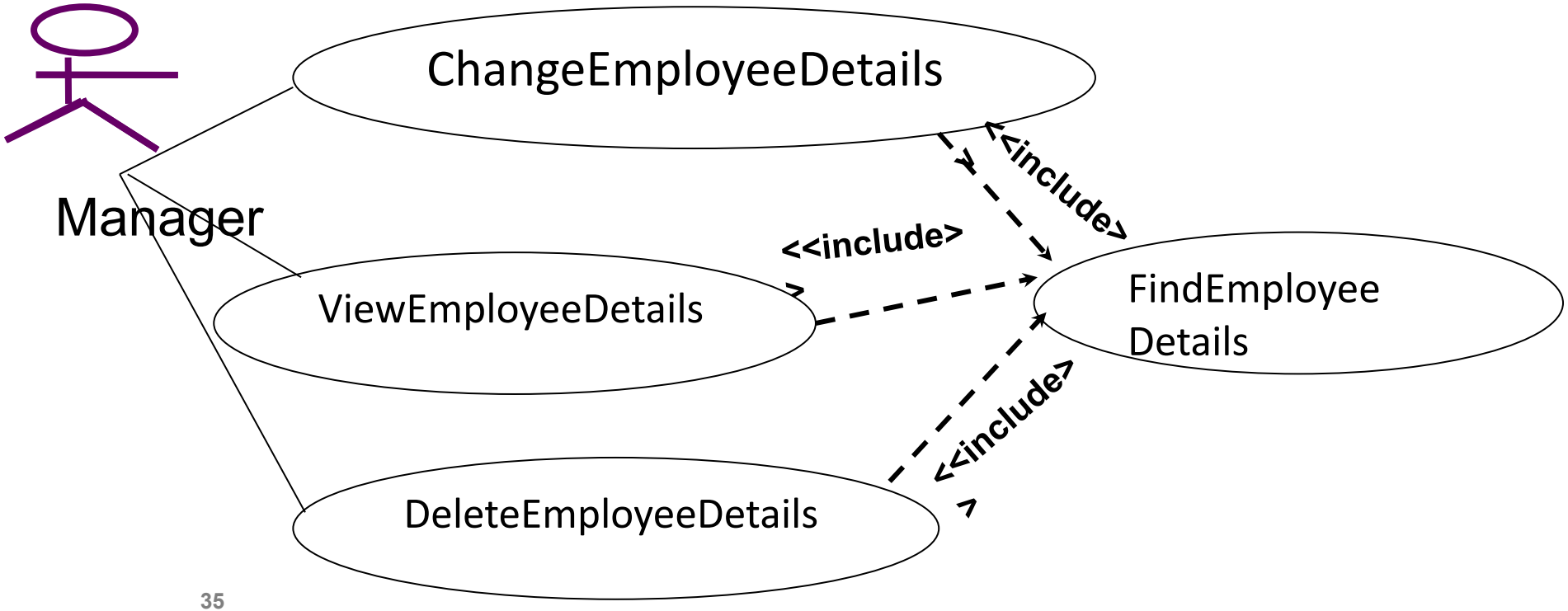
Use Case Diagram

- Relationships **<<include>>**
  - Include
    - The base use case explicitly incorporates the behavior of another use case.
    - The relationship between the abstract use case and use case that uses it.

Abstract use case: a use case that reduces redundancy among two / more other use cases by combining the common steps found in those cases.

Use Case Diagram

- Relationships
  - Uses (or include)
    - Another use case uses or includes the abstract use case.



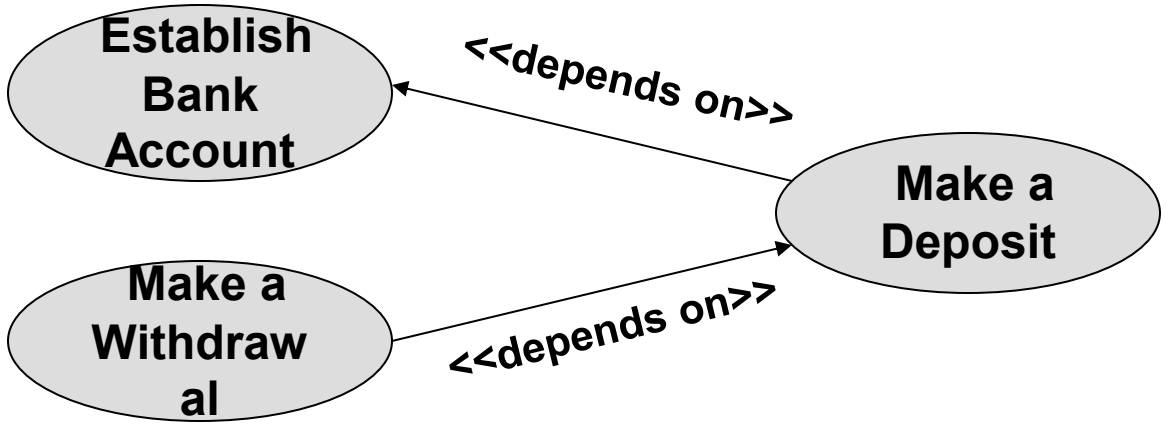
Use Case Diagram

- Relationships
  - Depends on
    - A relationship between use cases indicating that one use case cannot be performed until another use case has been performed.
      - e. g. banking business – use case ‘ Make a Withdrawal’ cannot be performed until the use case ‘Make a Deposit’ has been executed.

<<depends on>>

Use Case Diagram

- Relationships
  - Depends on

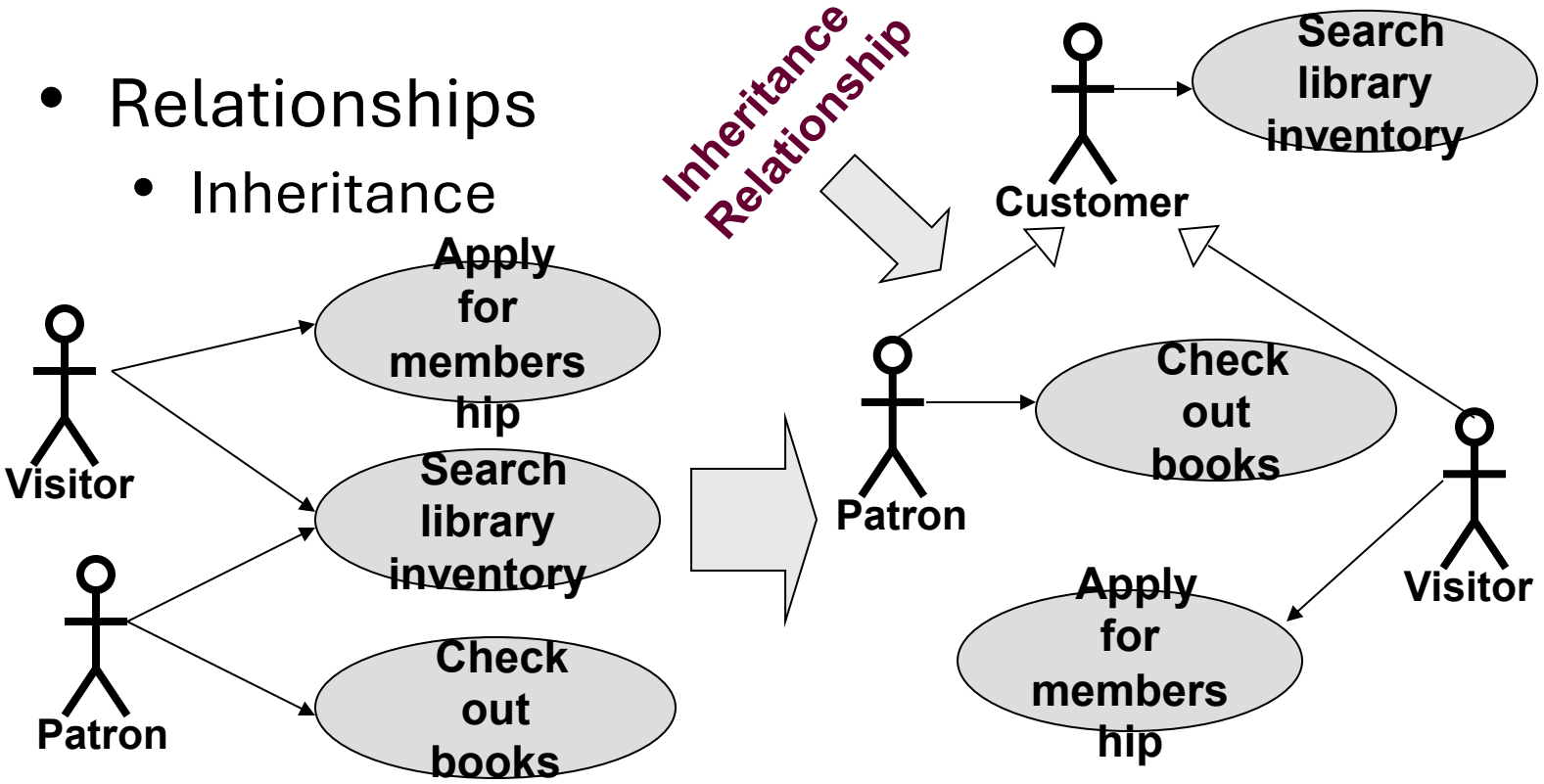


Use Case Diagram

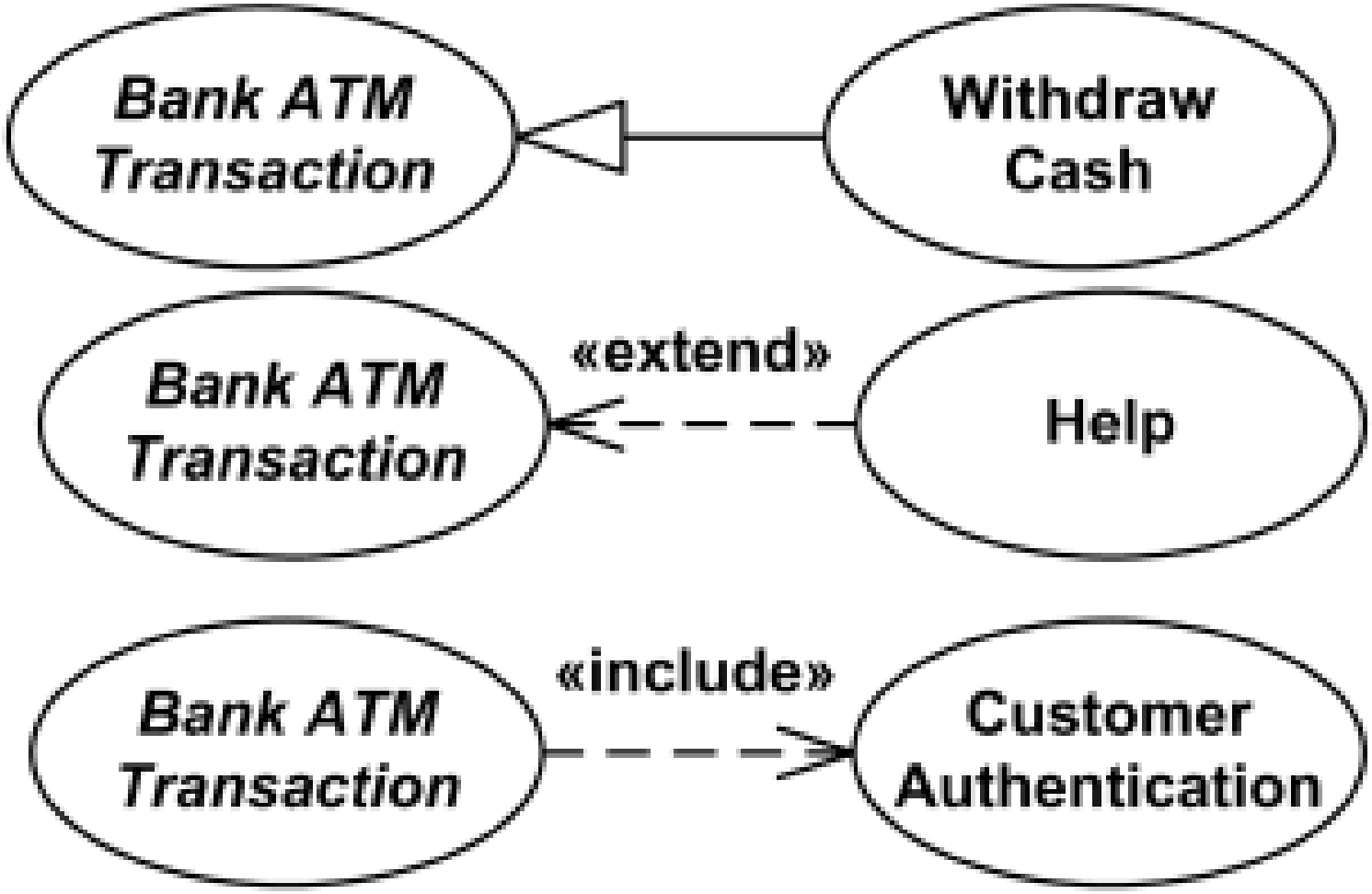
- Relationships
  - Generalization
    - A relationship between actors created to simplify the drawing when an abstract actor inherits the role of multiple real actors

Use Case Diagram

- Relationships
  - Inheritance



Include, extend and generalization Compared



Use Case Diagram

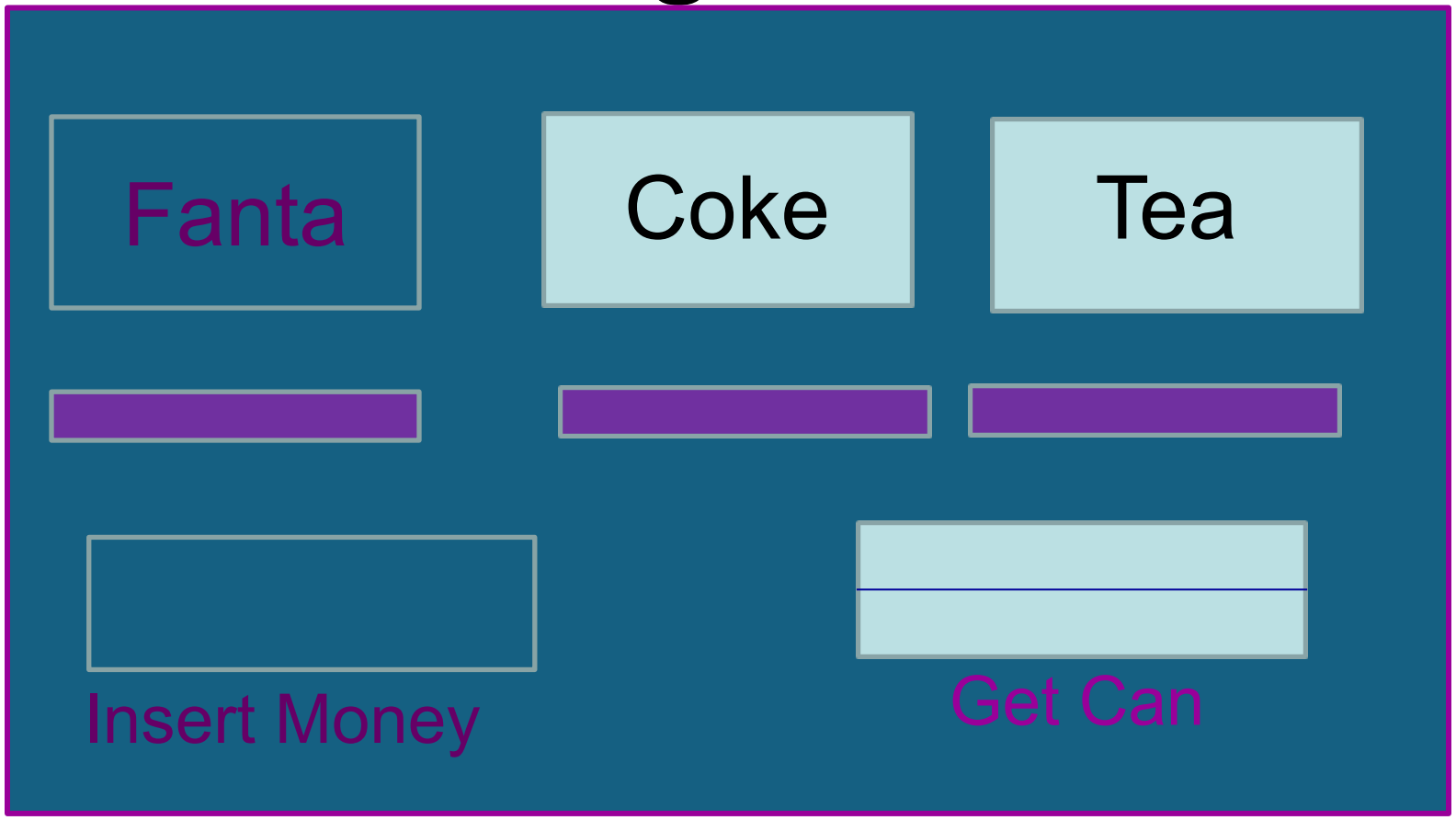
- System Boundary
  - A Box drawn around the Use Cases to denote the boundary of the system being modeled.
  - Refer to as the **Subject** in UML 2.

Process of Requirements Use Case Modelling

- Objective
  - analyze enough requirements information to prepare a model that communicates what is required from a user perspective
  - It is free of specific details about how the system will be built and implemented.
- Steps required to produce a model
  - Identify business actors
  - Identify business requirements use cases
  - Construct use-case model diagram
  - Document business requirements use-case narratives

  Use Case model should also identify the System Boundary

Use Case Modelling : Exercise
Vending Machine



Identify **Actors**, **Use Cases** and Draw a **Use Case Diagram**

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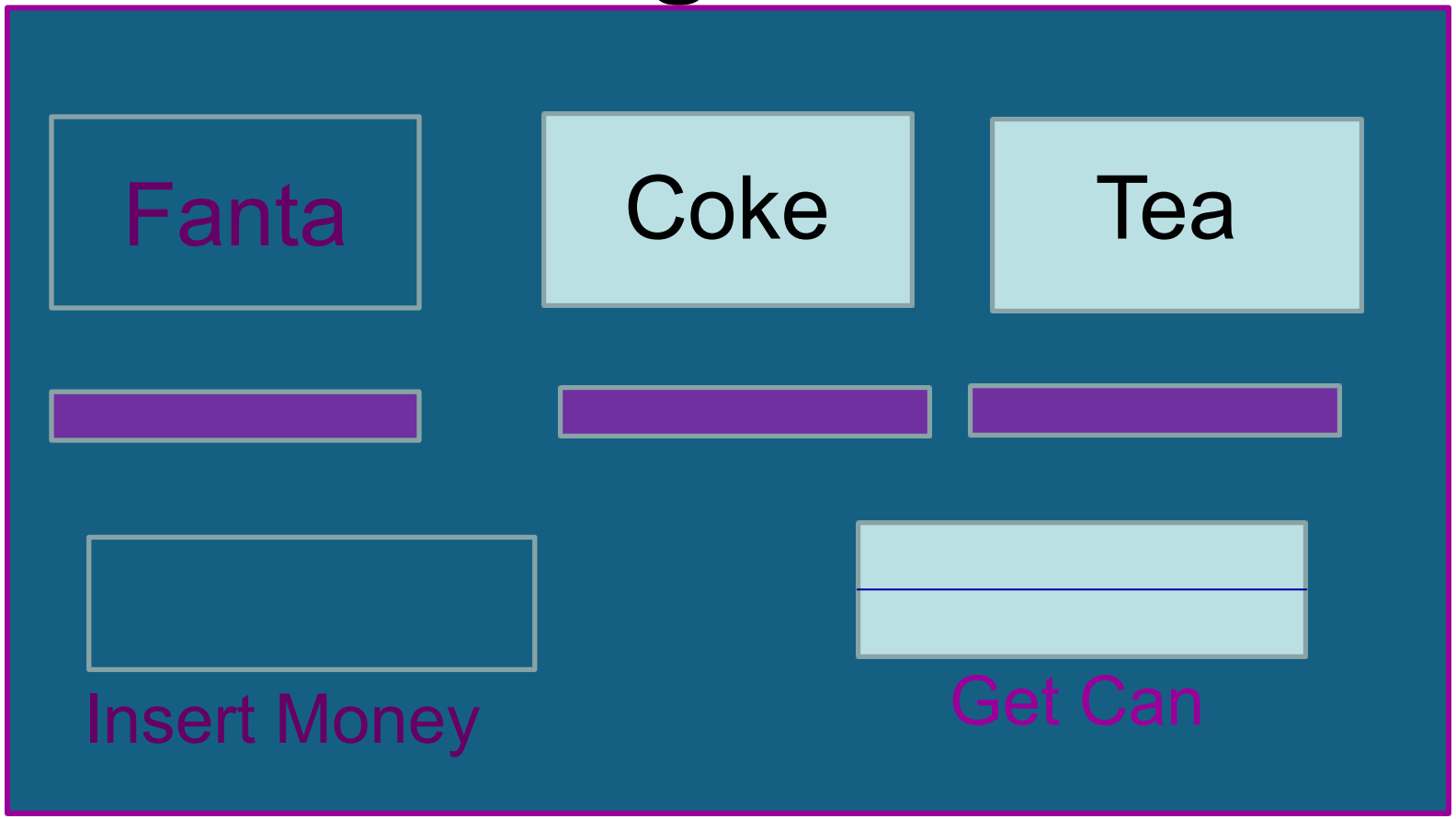
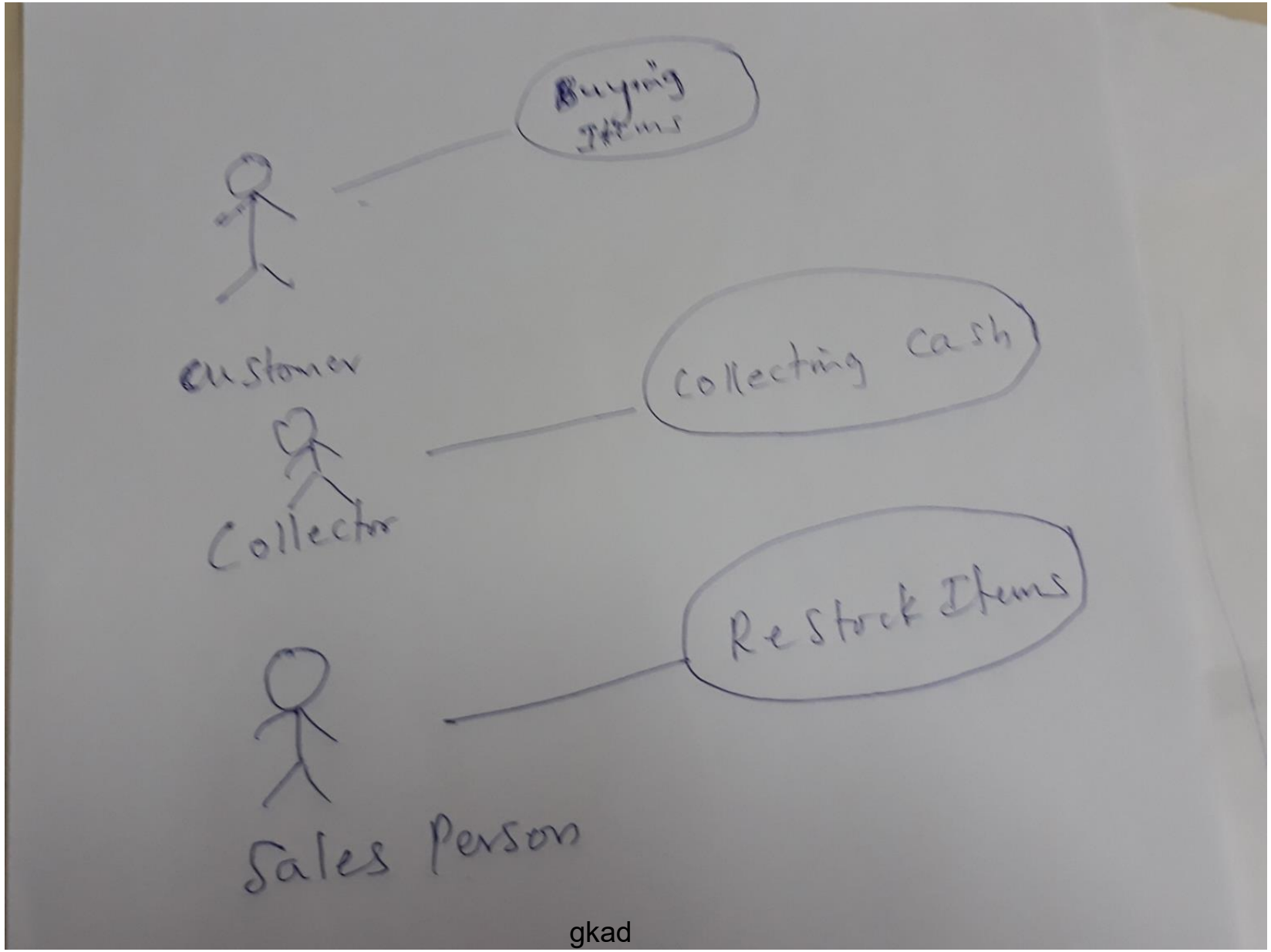
Use Case Modelling : Exercise
Vending Machine

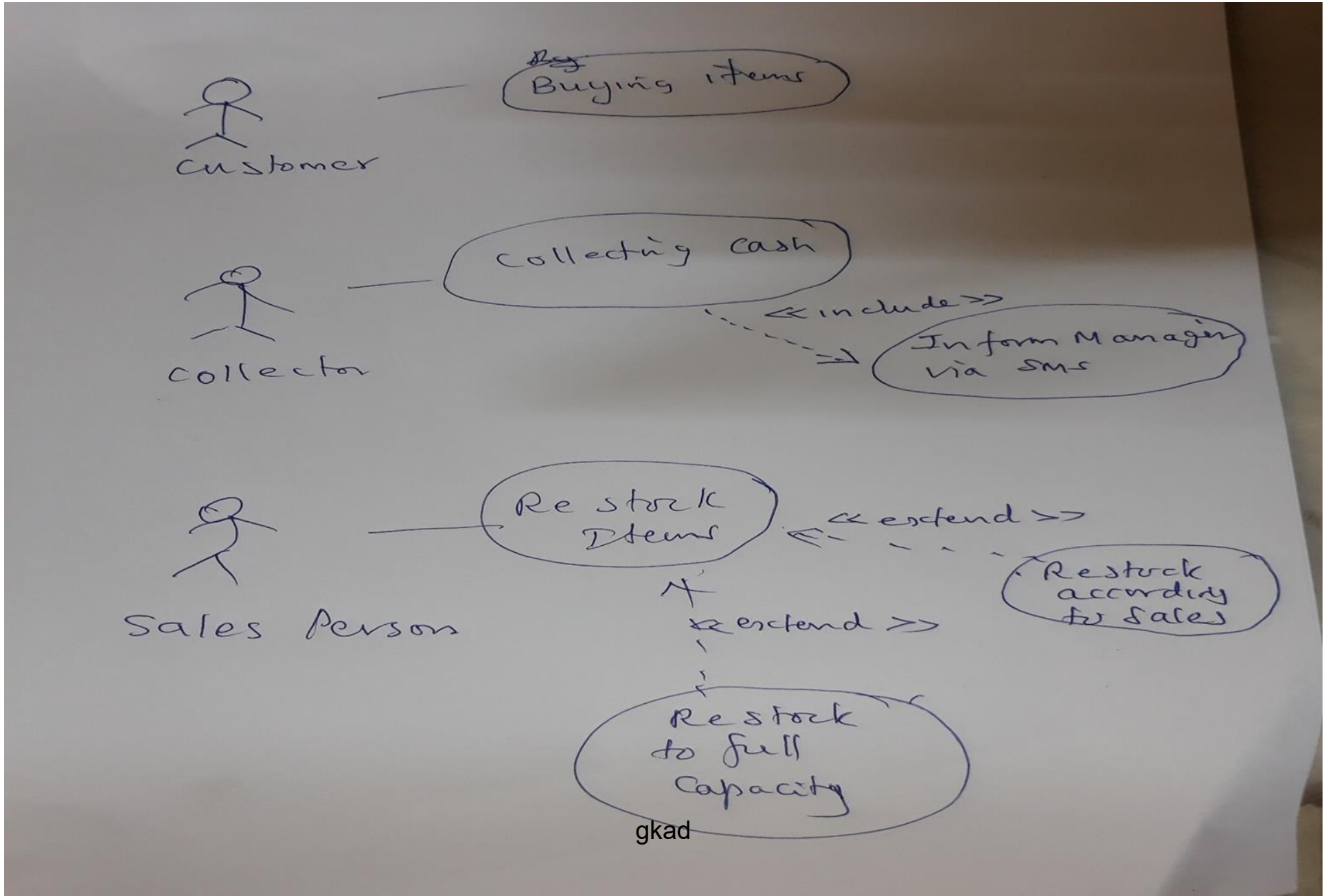


Identify **Actors**, **Use Cases** and Draw a **Use Case Diagram**

A Solution


Buying Items
Customer
Collecting Cash
Collector
Restock Items
Sales Person

A Solution Conti…



The Process of Requirements Use-Case Modeling

- Objective is to elicit and analyze enough requirements information to prepare a model that:
  - Communicates what is required from a user perspective.
  - Is free of specific details about how system will be implemented.
- To effectively estimate and schedule project, may need to include preliminary implementation assumptions.
- Steps
  1. Identify business actors.
  2. Identify business use cases.
  3. Construct use-case model diagram.
  4. Documents business requirements use-case narratives.

- **Step 1: Identify business actors**
  - Concentrate on how the system will be used and not how it will be built.
  - Actors should be named with a noun or noun phase.

- Where do you look for potential actors?
  - Existing system documentation and user manuals.
  - Minutes of project meetings and workshops.
  - Existing requirements documents

  In UML 2, actors may also represents other Subjects- gives you a way to link different Use Case models.

- When looking for actors, ask the following questions:
  - Who / What provides inputs to the system?
  - Who / What receives outputs from the system?
  - Are interfaces required to other systems?
  - Are there any events that are automatically triggered at a predetermined time?
  - Who will support and maintain information in the system?

- When looking for actors, ask the following questions:
  - Who is interested in a certain requirement?
  - Who will benefit from the use of the system?
  - Who will supply the system with this information, use this information, and remove this information?
  - Does one person play several different roles?
  - Does the system use an external resource?
  - Do several people play the same roles?

By focusing first on **actors**,

- you concentrate on how the system will be used instead of how it will be built.
- helps refine and further define the scope and boundaries of the system. Eg. Omit supplier details in a library
- Also, by first identifying actors you find candidates to interview and observe so you can develop and validate the use cases.

Sample List of Actors

	Term	Synonym	Description
1.	Potential member		An individual or corporation that submits a subscription order in order to join the club.
2.	Club member	Member	An individual or corporation that has joined the club via an agreement.
3.	Past member	Inactive member	A type of member that has fulfilled the agreement obligation but has not placed an order within the last six months but is still in good standing.
4.	Marketing		Organization responsible for creating promotion and subscription programs and generating sales for the company.
5.	Member services		Organization responsible for providing point of contact for SoundStage Entertainment customers in terms of agreements and orders.
6.	Distribution center	Warehouse	Entity that houses and maintains SoundStage Entertainment product inventory and processes customer shipments and returns.
7.	Accounts receivable		Organization responsible for processing customer payments and billing as well as maintaining customer account information.
8.	Time		Actor concept responsible for triggering temporal events.

- **Step 2: Identify Business Requirements Use Cases**
  - Business requirements use case capture the interactions between a user and the business/system **free of technology and implementation details**
  - Use case describes how a real-world actor interacts with the business/system

- When looking for use-cases, ask the following questions:
  - What are the main tasks of the actor?
  - What information does the actor need from the system?
  - What information does the actor provide to the system?
  - Does the system need to inform the actor of any changes or events that have occurred?
  - Does the actor need to inform the system of any changes or events that have occurred?

Process of Requirements Use Case Modelling

- When looking for use-cases, ask the following questions:
  - Will any actor create, store, change, remove, or read this information?
  - What use cases will create, store, change, remove, or read this information?
  - Can all functional requirements be performed by the use cases?
- Use cases are named with a verb phrase specifying the goal of an actor.
  Eg. Member Registration

Sample Use-Case Glossary

Use-Case Name	Use-Case Description	Participating Actors and Roles
Submit Subscription Order	This use case describes the event of a potential member requesting to join the club by subscribing. ("Take any 12 CDs for one penny and agree to buy 4 more at regular prices within two years.")	• Potential member (primary business) • Distribution Center (external receiver)
Submit Subscription Renewal Order	This use case describes the event of a past member requesting to rejoin the club by subscribing. ("Take any 12 CDs for one penny and agree to buy 4 more at regular prices within two years.")	• Past member (primary business) • Distribution Center (external receiver)
Submit Member Profile Changes	This use case describes the event of a club member submitting changes to his or her profile for such things as postal address, e-mail address, privacy codes, and order preferences.	• Club member (primary business)
Place New Order	This use case describes the event of a club member submitting an order for SoundStage products.	• Club member (primary business) • Distribution Center (external receiver) • Accounts Payable/Receivable (external server)

continued

Sample Use-Case Glossary (cont.)

Revise Order	This use case describes the event of a club member revising an order previously placed. (Order must not have shipped.)	• Club member (primary business) • Distribution Center (external receiver) • Accounts Payable/Receivable (external server)
Cancel Order	This use case describes the event of a club member canceling an order previously placed. (Order must not have shipped.)	• Club member (primary business) • Distribution Center (external receiver) • Accounts Payable/Receivable (external server)
Make Product Inquiry	This use case describes the event of a club member viewing products for possible purchase. (Driven by web access requirement.)	• Club member (primary business)
Make Purchase History Inquiry	This use case describes the event of a club member viewing her or his purchasing history. (Three-year time limit.)	• Club member (primary business)

continued

Sample Use-Case Glossary (cont.)

Establish New Member Subscription Program	This use case describes the event of the marketing department establishing a new membership subscription plan to entice new members	• Marketing (primary business)
Submit Subscription Program Changes	This use case describes the event of the marketing department changing a subscription plan for club members (e.g., extending the fulfillment period).	• Marketing (primary business)
Establish Past Member Resubscription Program	This use case describes the event of the marketing department establishing a resubscription plan to lure back former members.	• Marketing (primary business)
Submit Member Profile Changes	This use case describes the event of the marketing department establishing a new promotion plan to entice active and inactive members to order the product. (Note: A promotion features specific titles, usually new, that the company is trying to sell at a special price. These promotions are integrated into a catalog sent (or communicated) to all members.)	• Marketing (primary business)
Revise Promotion	This use case describes the event of the marketing department revising a promotion.	• Marketing (primary business)
Generate Daily 10-30-60-Day Default Agreement Report	This use case describes the event of a report that is generated on a daily basis to list the members who have not fulfilled their agreement by purchasing the required number of products outlined when they subscribed. This report is sorted by members who are 10 days past due, 30 days past due, and 60 days past due.	• Time (initiating actor) • Member Services (primary* — external receiver)

- **Step 3: Construct Use-Case Model Diagram**
  - Once use-cases and actors have been identified, a use-case model diagram can be used to show the system scope and boundaries.
  - If you have several sub systems, you may draw several use case model diagrams.

  Eg. Library System, Registration System, Examination System

Use-Case Modelling

- **Step 4: Document Business Requirements Use-case Narratives**
  - When preparing narratives,
    - First document them at a high level (can get a quick understanding of the events and magnitude of the system)
    - Then expand each use case separately to a fully documented business requirement narrative.

  For more information refer pg 256 of Ref 3

e.g. High Level Version of a use-case narrative

Author (s): ----------

Date:----------

Version: ----------

Use-Case Name:		Use-Case Type Business Requirements:
Use-Case ID:		
Priority:		
Source:		
Primary Business Actor:		
Other Participating Actors:		
Other Interested Stakeholders:		
Description:		

Importance of the Use Case – typically high , medium , low

e.g. High Level Version of a use-case narrative

Author (s): ----------

Date:----------

Version: ----------

Use-Case Name:		Use-Case Type Business Requirements:
Use-Case ID:		
Priority:		
Source:		
Primary Business Ac		
Other Participating Actors:		
Other Interested Stakeholders:		
Description:		

Entity that triggers the creation of the Use Case. Eg. Document

e.g. High Level Version of a use-case narrative

Author (s): ----------

Date:----------

Version: ----------

Use-Case Name:		Use-Case Type Business Requirements:
Use-Case ID:		
Priority:		
Source:		
Primary Business Actor:		
Other Participating Actors:		
Other Interested Stakeholders:		
Description:		

Who benefits from the use case

e.g. High Level Version of a use-case narrative

Author (s): ----------

Date:----------

Version: ----------

Use-Case Name:		Use-Case Type Business Requirements:
Use-Case ID:		
Priority:		
Source:		
Primary Business Actor:		
Other Participating Actors:		
Other Interested Stakeholders:		
Description:		

Facilitating Actors

e.g. High Level Version of a use-case narrative

Author (s): ----------

Date:----------

Version: ----------

Use-Case Name:		Use-Case Type Business Requirements:
Use-Case ID:		
Priority:		
Source:		
Primary Business Actor:		
Other Participating Actors:		
Other Interested Stakeholders:		
Description:		

General understanding of problem domain and scope

In brief

Sample High-Level Use-Case Narrative

Member Services System

Author (s): 1 **Date:** 2
Version: 3

Use-Case Name:	Place New Order 4	**Use-Case Type**
Use-Case ID:	MSS-BUC002.00 6	**Business Requirements:** ☑ 5
Priority:	High 7	
Source:	Requirement — MSS-R1.00 8	
Primary Business Actor:	Club member 9	
Other Participating Actors:	• Warehouse (external receiver) • Accounts Receivable (external server) 10	
Other Interested Stakeholders:	11 • Marketing — Interested in sales activity in order to plan new promotions. • Procurement — Interested in sales activity in order to replenish inventory. • Management — Interested in order activity in order to evaluate company performance and customer (member) satisfaction.	
Description: 12	This use case describes the event of a club member submitting a new order for SoundStage products. The member's demographic information as well as his or her account standing is validated. Once the products are verified as being in stock, a packing order is sent to the warehouse for it to prepare the shipment. For any product not in stock, a back order is created. On completion, the member will be sent an order confirmation.	

e.g. Expanded Version of a use-case narrative

More details such as
- Preconditions
- Trigger
- Typical Course of Event
- Alternate Courses
- Post conditions

etc. are included.
(for more information Refer pg258 of Ref 1)

Typically another Use Case that must be previously executed.

e.g. Expanded Version of a use-case narrative

More details such as

- Preconditions
- Trigger
- Typical Course of Even
- Alternate Courses
- Post conditions

etc. are included.

(for more information Refer pg258 of Ref 1)

Time receiving a cheque.

e.g. Expanded Version of a use-case narrative

More details such as
- Preconditions
- Trigger
- Typical Course of Events
- Alternate Courses
- Post conditions

etc. are included.
(for more information Refer pg258 of Ref 1)

eg. Borrowing :
checkMember,
checkOverdue,
CheckOverLimit,
checkCopyBorrowable
, Confirm Borrowing

e.g. Expanded Version of a use-case narrative

More details such as
- Preconditions
- Trigger
- Typical Course of Events
- Alternate Courses
- Post conditions

etc. are included.
(for more information Refer pg258 of Ref 1)

Errors, Confirm Messages

e.g. Expanded Version of a use-case narrative

More details such as
- Preconditions
- Trigger
- Typical Course of Events
- Alternate Courses
- Post conditions

etc. are included.
(for more information Refer pg259 of Ref 3)

Receipt Delivered to the Customer

Sample Expanded Use-Case Narrative

Member Services System

Author (s): _____ **Date:** _____

Version:

Use-Case Name:	Place New Order	**Use-Case Type** **Business Requirements:** ☑
Use-Case ID:	MSS-BUC002.00	
Priority:	High	
Source:	Requirement — MSS-R1.00	
Primary Business Actor:	Club member	
Other Participating Actors:	• Warehouse (external receiver) • Accounts Receivable (external server)	
Other Interested Stakeholders:	• Marketing — Interested in sales activity in order to plan new promotions. • Procurement — Interested in sales activity in order to replenish inventory. • Management — Interested in order activity in order to evaluate company performance and customer (member) satisfaction.	
Description:	This use case describes the event of a club member submitting a new order for SoundStage products. The member's demographic information as well as his or her account standing is validated. Once the products are verified as being in stock, a packing order is sent to the warehouse for it to prepare the shipment. For any product not in stock, a back order is created. On completion, the member will be sent an order confirmation.	
Precondition: 1	The party (individual or company) submitting the order must be a member.	
Trigger: 2	This use case is initiated when a new order is submitted.	

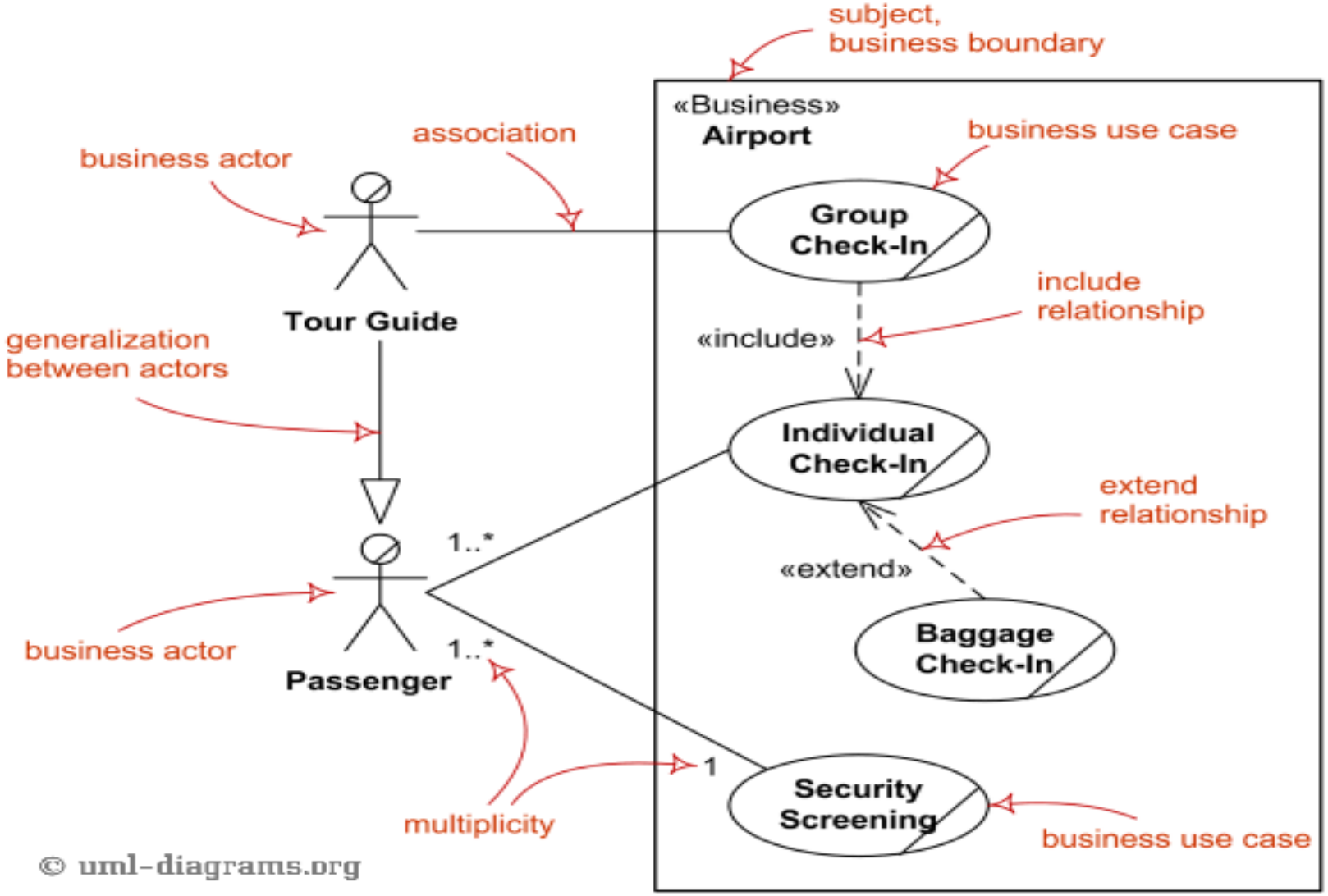
Expanded Use-Case Narrative (cont)

Typical Course of Events: 3	Actor Action	System Response
	Step 1: The club member provides his or her demographic information as well as order and payment information.	**Step 2:** The system responds by verifying that all required information has been provided. **Step 3:** The system verifies the club member's demographic information against what has been previously recorded. **Step 4:** For each product ordered, the system validates the product identity. **Step 5:** For each product ordered, the system verifies the product availability. **Step 6:** For each available product, the system determines the price to be charged to the club member. **Step 7:** Once all ordered products are processed, the system determines the total cost of the order. **Step 8:** The system checks the status of the club member's account. **Step 9:** The system validates the club member's payment if provided. **Step 10:** The system records the order information and then releases the order to the appropriate distribution center (warehouse) to be filled. **Step 10:** Once the order is processed, the system generates an order confirmation and sends it to the club member.

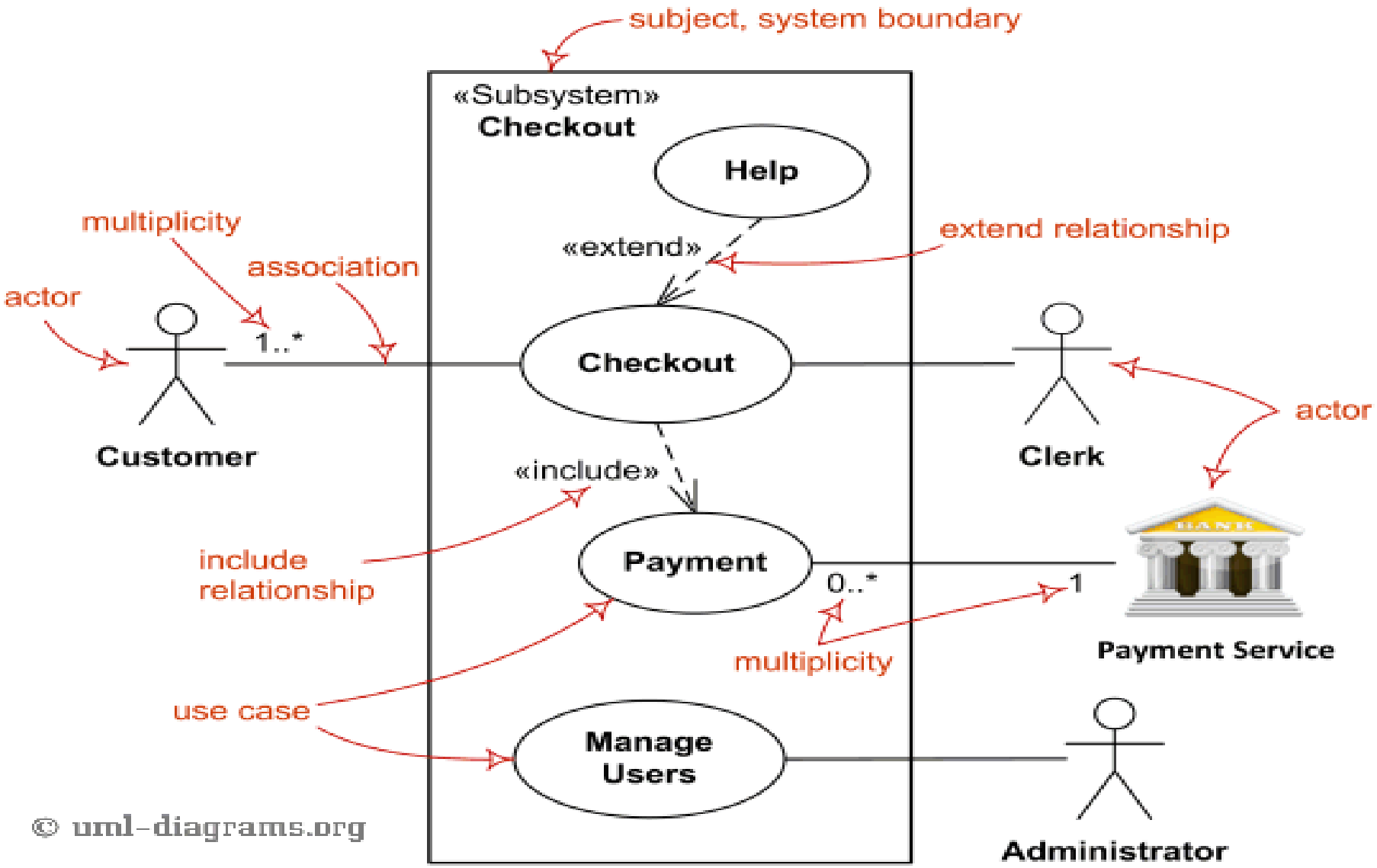
Expanded Use-Case Narrative (cont)

Alternate Courses: 4	**Alt-Step 2:** The club member has not provided all the information necessary to process the order. The club member is notified of the discrepancy and prompted to resubmit. **Alt-Step 3:** If the club member information provided is different from what was previously recorded, verify what was recorded is current, then update the club member information accordingly. **Alt-Step 4:** If the product information the club member provided does not match any of SoundStage's products, notify the club member of the discrepancy and request clarification. **Alt-Step 5:** If the quantity ordered of the product is not available, a back order is created. **Alt-Step 8:** If the status of the club member's account is not in good standing, record the order information and place it in hold status. Notify the club member of the account status and the reason the order is being held. Terminate use case. **Alt-Step 9:** If the payment the club member provided (credit card) cannot be validated, notify the club member and request an alternative means of payment. If the club member cannot provide an alternate means, cancel the order and terminate the use case.
Conclusion: 5	This use case concludes when the club member receives a confirmation of the order.
Postcondition: 6	The order has been recorded and if the ordered products were available, they were released. For any product not available a back order has been created.
Business Rules: 7	• The club member responding to a promotion or a member using credits may affect the price of each ordered item. • Cash or checks will not be accepted with the orders. If provided, they will be returned to the club member. • The club member is billed for products only when they are shipped.
Implementation Constraints and Specifications: 8	• GUI to be provided for Member Services associate, and web screen to be provided for club member.
Assumptions: 9	Procurement will be notified of back orders by a daily report (separate use case).
Open Issues: 10	1. Need to determine how distribution centers are assigned.

Business Use Case Model in RUP (Rational Unified Process)



System Use Case Diagram



Use-Cases and Project Management

- Use-case model can be used to drive the entire system development effort.
- Once the use-case model is complete the project manager or system analyst uses the model to plan the build cycles of the project.
- To determine the importance of the use-cases, the project manager or system analyst will complete **a use-case ranking and evaluation matrix** and construct a **use case dependency diagram** with inputs from the stake holders and the development team.

Use-Cases and Project Management

- **Ranking and Evaluating Use Cases**
  - Most important use-cases should be developed first
  - ***Use-case ranking and priority matrix*** is a tool used to evaluate use cases and determine their priority

Use-Cases and Project Management

- ***Use-case ranking and priority matrix***
  - Completed with input from the stakeholders and the development team
  - Adapted from Craig Larman's work
  - Evaluates use-cases on a scale of 1-5 against six criteria

    1. **Significant impact on the architectural design**
    2. **Easy to implement but contains significant functionality**
    3. **Includes risky, time-critical, or complex functions**
    4. **Involves significant research or new or risky technology**
    5. **Includes primary functions**
    6. **Will increase revenue or decrease cost**

Use-Cases and Project Management

Criteria

Use-case ranking and priority matrix : e.g.

Use Case Name	Ranking Criteria, 1 to 6						Total Score	Priority	Build Cycle
	1	2	3	4	5	6			
Submit subscription order	5	5	5	4	5	5	29	High	1
Place new order	4	4	5	4	5	5	27	High	2
Make product inquiry	1	1	1	1	1	1	6	Low	3
Establish new member subscription program	4	5	5	3	5	5	27	High	1
Revise order	2	2	3	3	4	4	18	Medium	2

Use-Cases and Project Management

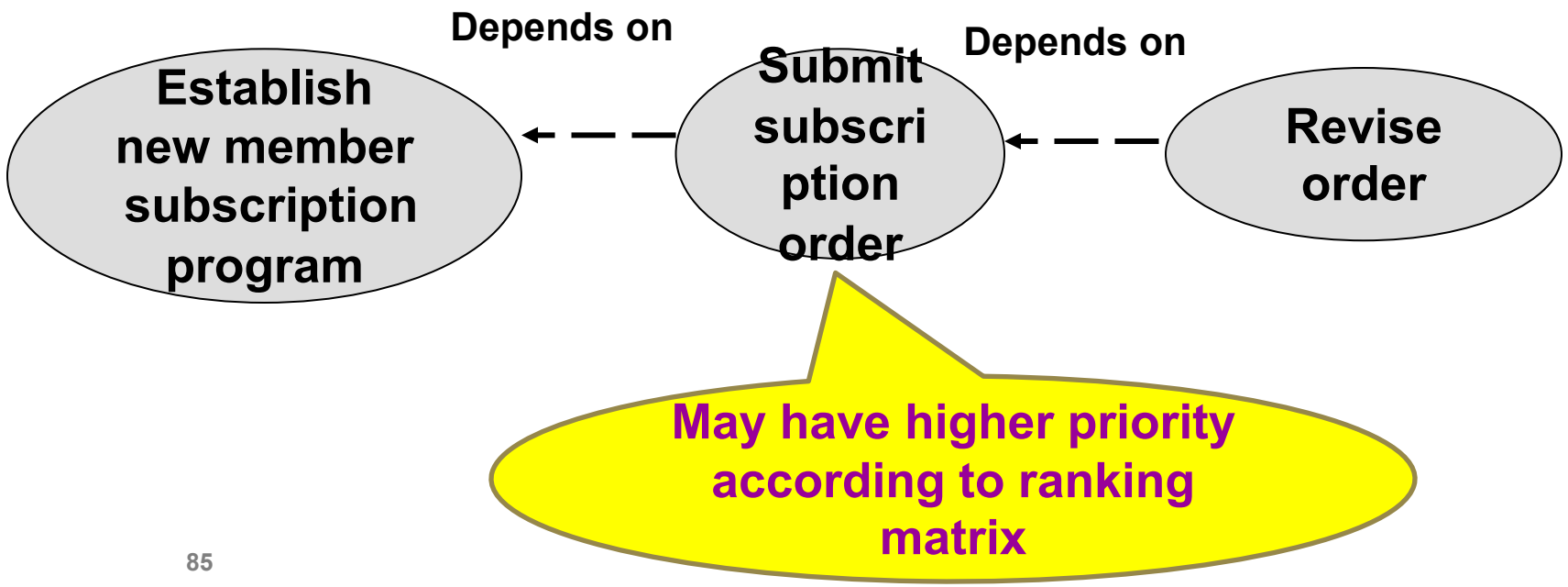
- **Identifying Use-Case dependencies**
  - Some use-cases may depend on other use-cases, with one use case leaving the system in a state that is a precondition for another use case.
  - Use-case dependency diagram is used to model such dependencies.

Use-Cases and Project Management

- **Use-case dependency diagram**
  - A graphical depiction of the dependencies among use-cases
  - Benefits
    - Enhances the understanding of system functionality
    - Helps to identify missing use-cases
    - Facilitate project management by depicting which use-cases are more critical and thus need to have a higher priority

Use-Cases and Project Management

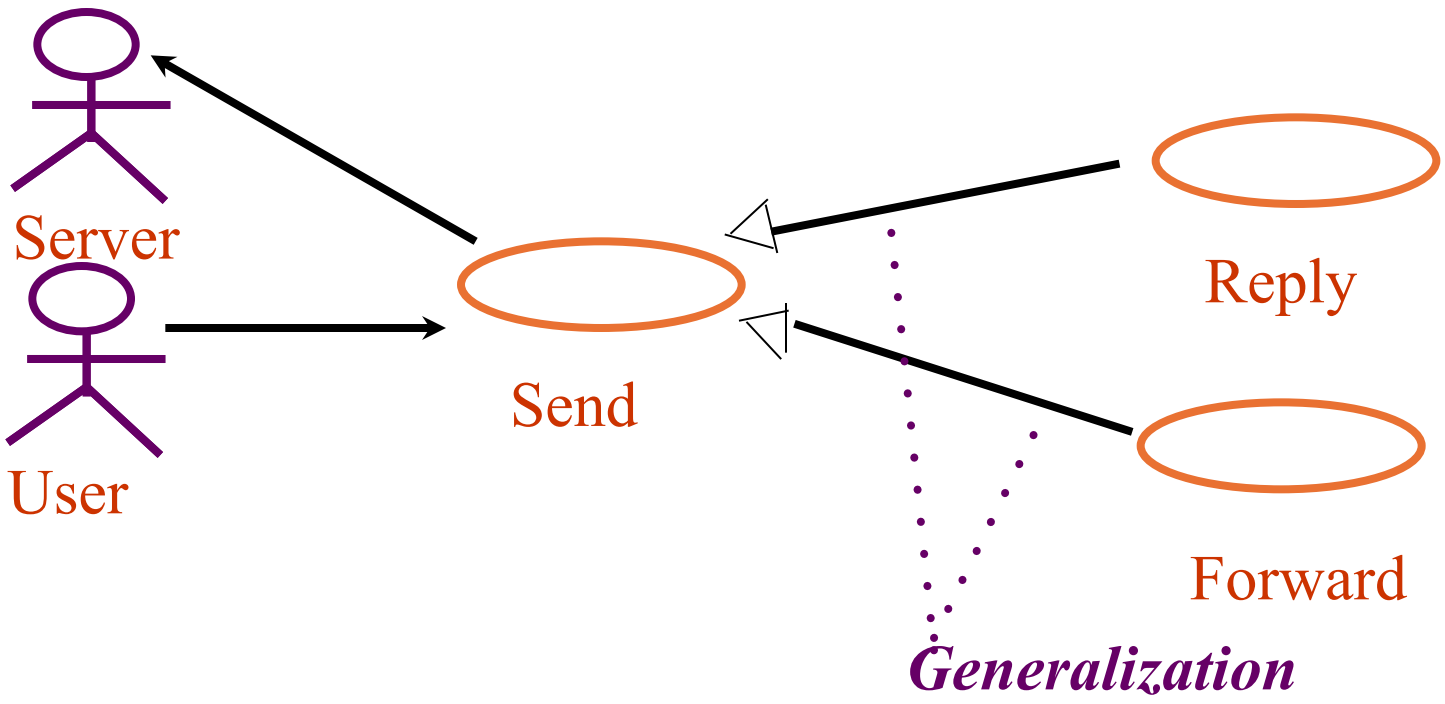
- Use-case dependency diagram: e.g.



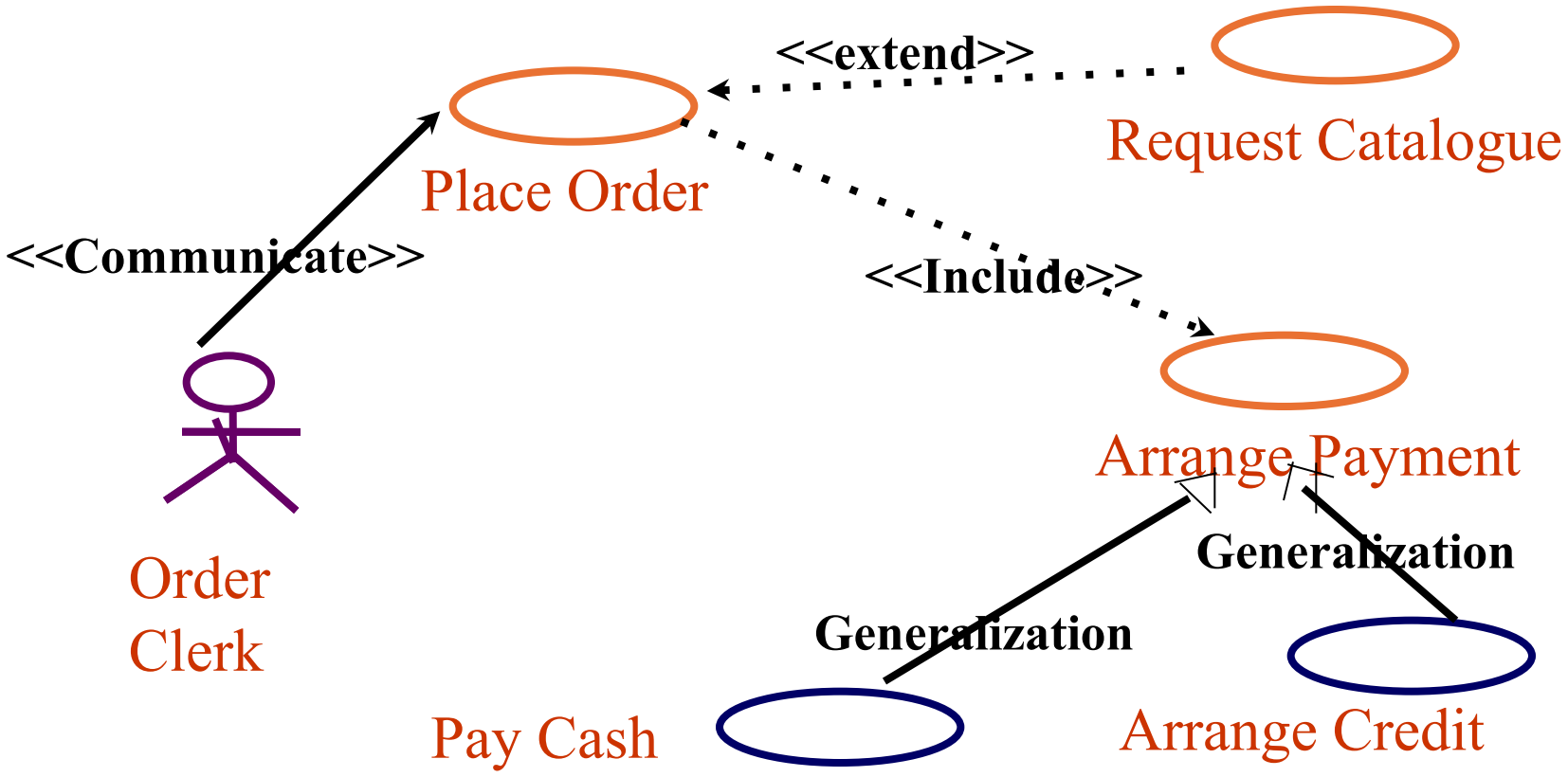
USE Case Modelling

Eg. USE CASE Diagram

User sending an E-mail message



Use Case diagram for an Order Processing Application.



General activities in performing Object Oriented Analysis

1. Modeling the functions of the system
2. Finding and Identifying the Business Objects
3. Organizing the Objects and Identifying the Relationships

1. Modeling the functions of the System

- Document functional system requirements using business requirements use cases
  - Use Cases documented to contain only general information about the business event.
  - Goal was to quickly document all of the business events (use cases) in order to define and validate requirements.
- Construct the Analysis Use Case Model

Construct the Analysis Use Case Model

- Evolve Requirements Use Case model into Analysis Use Case Model by performing the following 5 steps:
  1. Identify, define and document new actors
  2. Identify, define and document new Use Cases
  3. Identify any reuse possibilities
  4. Refine the Use case model diagram
  5. Document System analysis use case narratives

Construct the Analysis Use Case Model

1. Identify, define and document new actors
   - eg. Consider Use Case *Place New Order*

   Customer can place the order using Internet (identified initially)

   Customer also can submit orders by e-mail (Identified later)

   Service associates *input the order details sent through e-mail*

Construct the Analysis Use Case Model

2. Identify, define and document new Use Cases
   - eg. New actor ***Service associates*** *need to a interact with the system – need to add a new Use Case*
3. Identify any reuse possibilities
   - Two users can share common steps like in the place order example
   - Can extract the common steps into a separate Use Case. <<***include***>>

Construct the Analysis Use Case Model

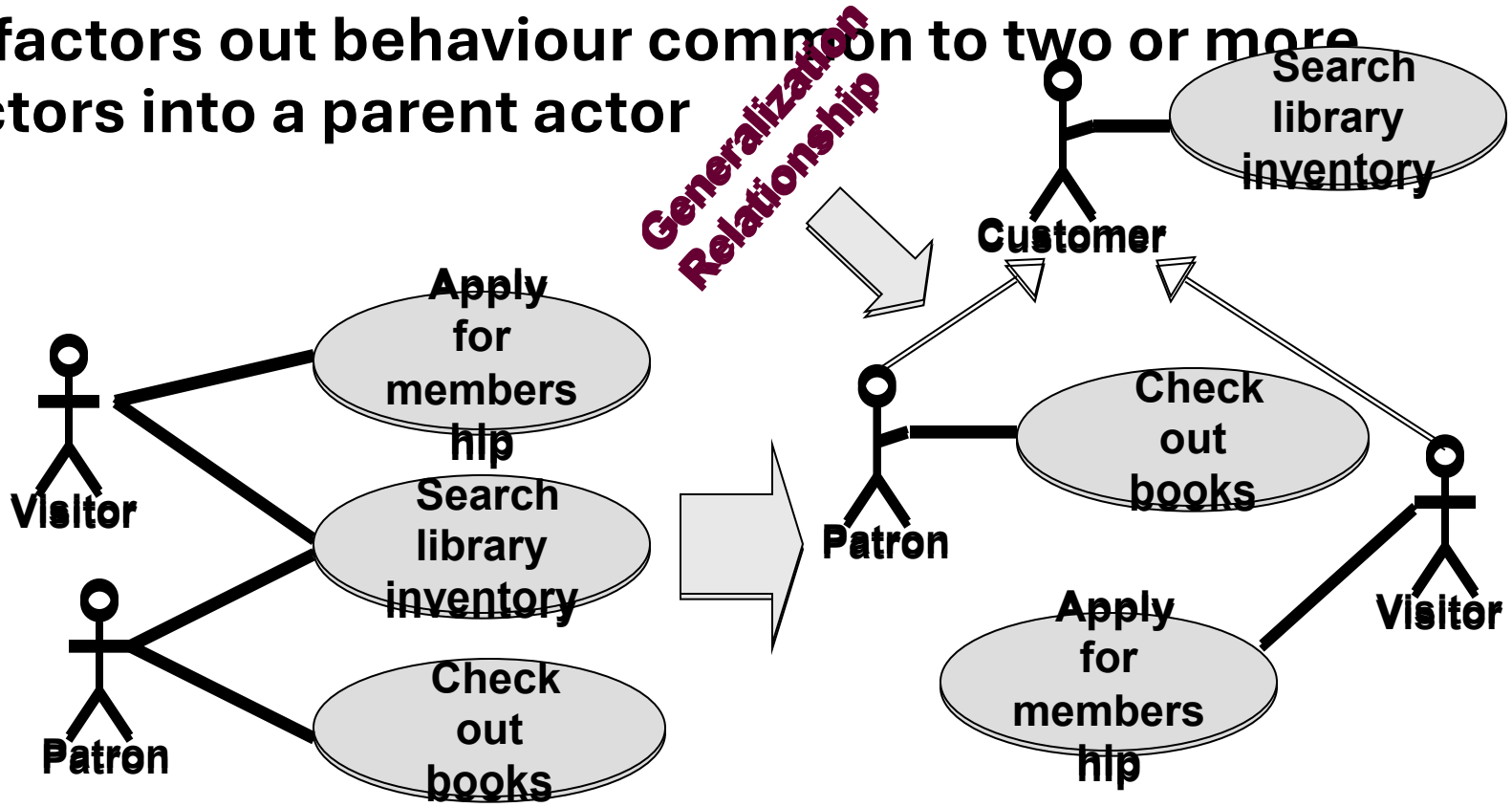
4. Refine the Use case model diagram (If necessary)
   - With the discovery of Actors and/or Use Cases , Use Case Model Diagram needs to be updated.
5. Document System analysis use case narratives.

   eg. Ref 3 page 386

Actor Generalization

We can simplify the model with Actor Generalization.

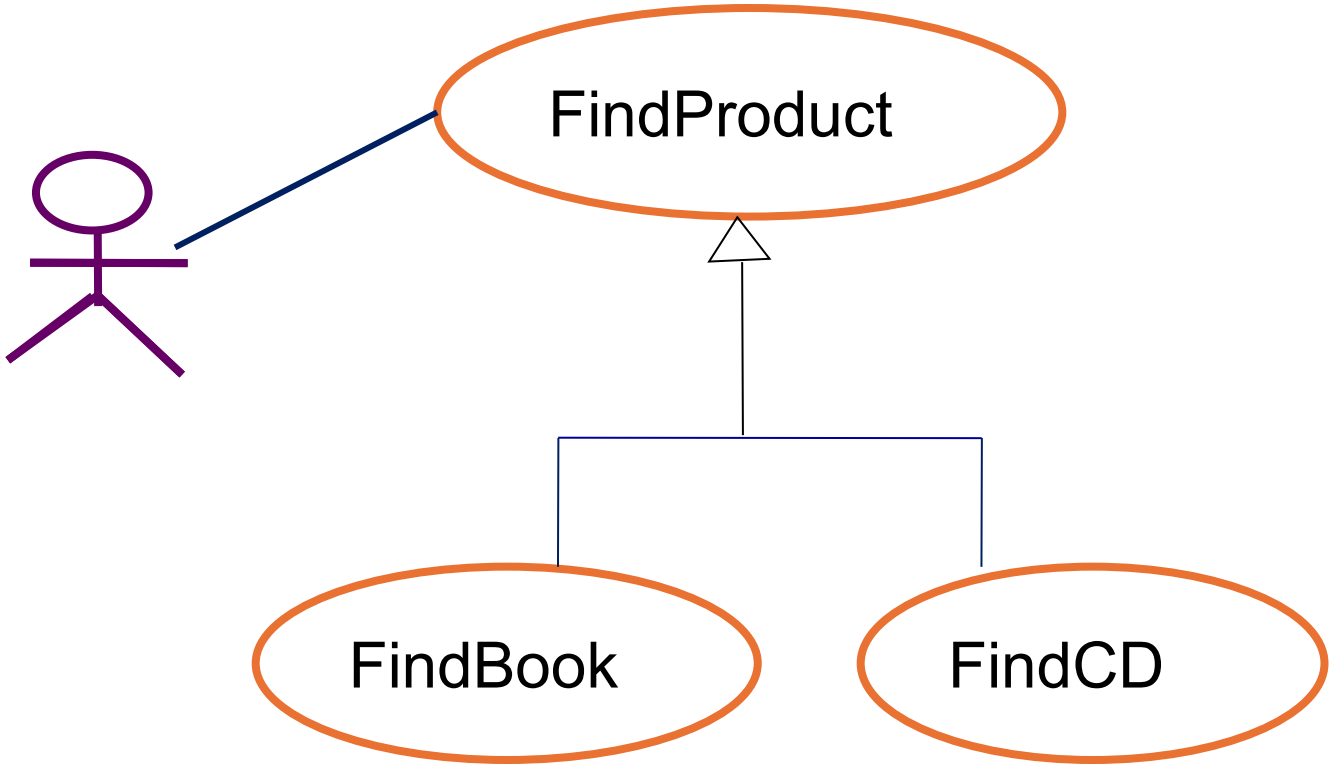
It factors out behaviour common to two or more actors into a parent actor



Use Case Generalization

- Factors out behaviour common to one or more use cases into a parent use case.
- In Use case generalization, the child use cases represent more specific forms of the parent. The children may:
  - Inherit features from their parent use case
  - Add new features
  - Override (change) inherited features

Use Case Generalization cont...



Specification for the parent Use case

Use case : Find Product ID- 6
Brief Description: The customer searches for product
Primary Actor : Customer
Secondary Actors : None
Preconditions : None
Main Flow : 1. The Customer selects **find product** 2. System asks for search criteria 3. The Customer enters the requested criteria 4. System search for products that match the Cust. criteria. 5. If the system find some matching products 5.1 The sys displays a list of matching products 6. Else 6.1 The sys tells customer “No matching Products”
Post Conditions, Alternative Flows : none

Specification for a child Use case

Use case : Find Book ID- 7 , Parent ID -6

Brief Description: The customer searches for a Book

Primary Actor , Secondary Actor: Customer

Preconditions : None

Main Flow :

1. The Customer selects **"find book"**
2. System asks for book search criteria (title, ISBN, etc)
3. The Customer enters the requested criteria
4. System srch for books that match the Cust. criteria.
5. If the system find some matching books
   5.1 The sys displays a details of matching books (max 5)
   5.2 While there are more books, cust. Is given the option to display the next page of books.
6. Else
   6.1 The sys tells customer "No matching Products"

Post Conditions, Alternative Flows : none

Overridden

Specification for a child Use case

Use case : Find Book ID- 7 , Parent ID -6

Brief Description: The customer searches for a Book

Pre: C

Pre

Main Flow :

1. The Customer selects “final book”
2. System asks for book search criteria (title, ISBN, etc)
3. The Customer enters the requested criteria
4. System srch for books that match the Cust. criteria.
5. If the system find some matching books
   5.1 The sys displays a details of matching books (max 5)
   5.2 While there are more books, cust. Is given the option to display the next page of books.
6. Else
   6.1 The sys tells customer “No matching Products”

Post Conditions, Alternative Flows : none

Add new features

Inherited without change

Use Case Generalization cont…

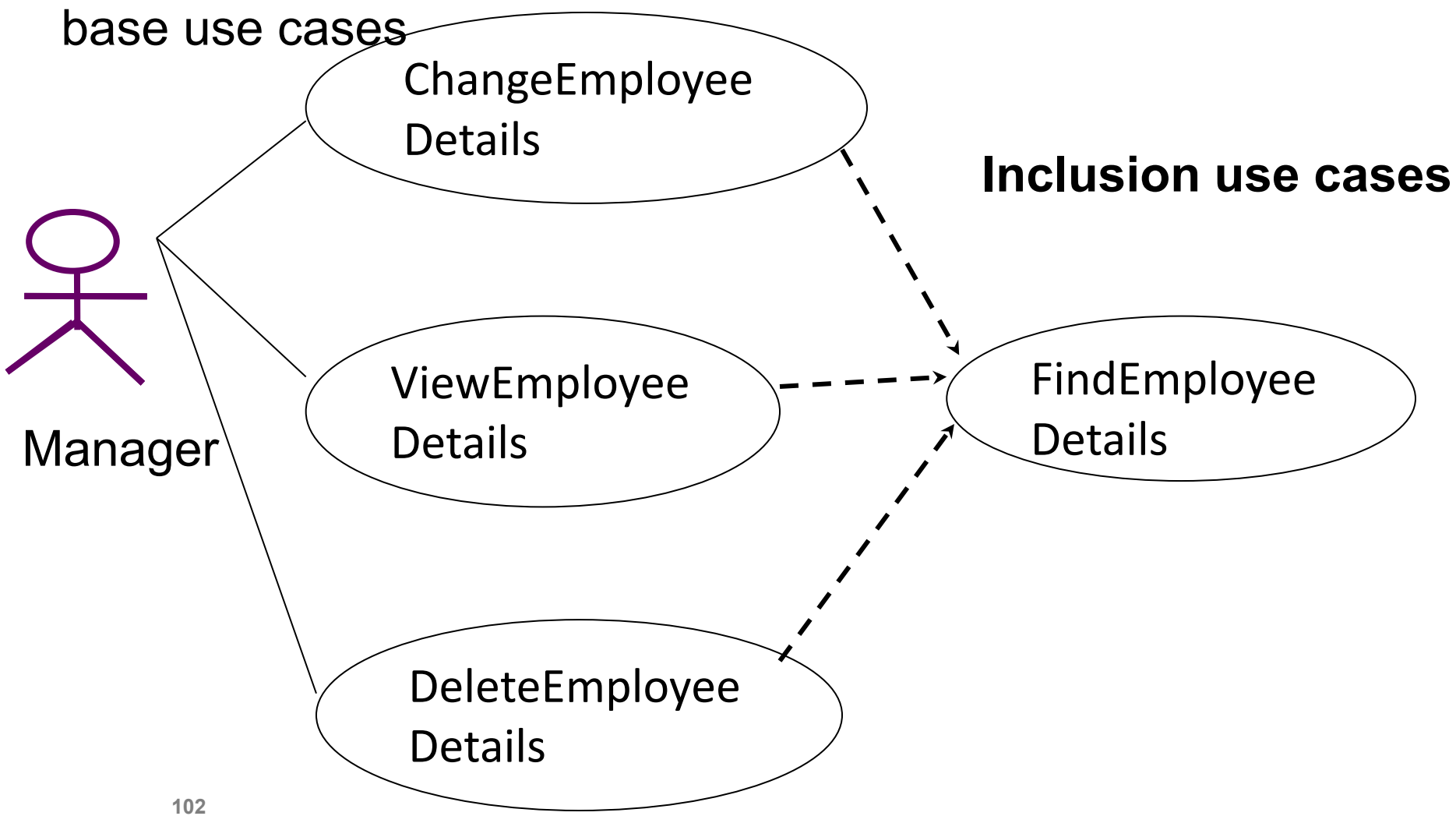
- Use case models are used to communicate with stakeholders.
- Users find it difficult to understand generalization features such as overriding etc.
- Most analyst restrict the parent use cases so that it doest not have any main flow, but only a brief description of its semantics.
- This approach makes use case generalization easy as the parent use case becomes an **abstract** use case.

Have a plain text summary of the high level behaviour that its children will implement.

Use Case Generalization cont…

- Since abstract use cases have missing or incomplete flow of events,
  - they will never be executed by the system
- You should only use Use Case generalization if it simplifies your use case model.

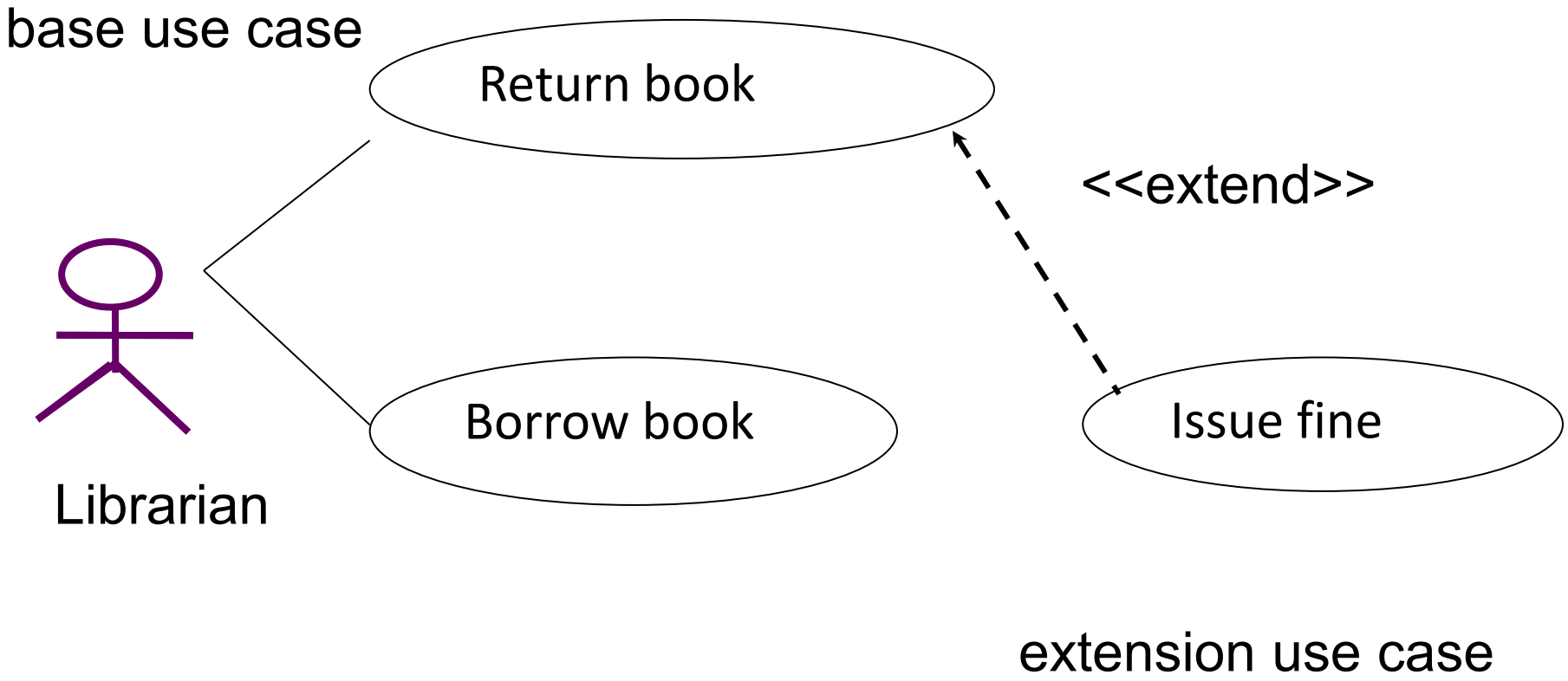
Example of <<include>> Relationship



Example of <<include>> Relationship cont..

- *Inclusion* use case supplies behavior to its base use case.
- The base use case executes until the point of inclusion is reached,
- Then execution passes over to the inclusion use case.
- When the inclusion use case finishes , the control return to the base use case again.

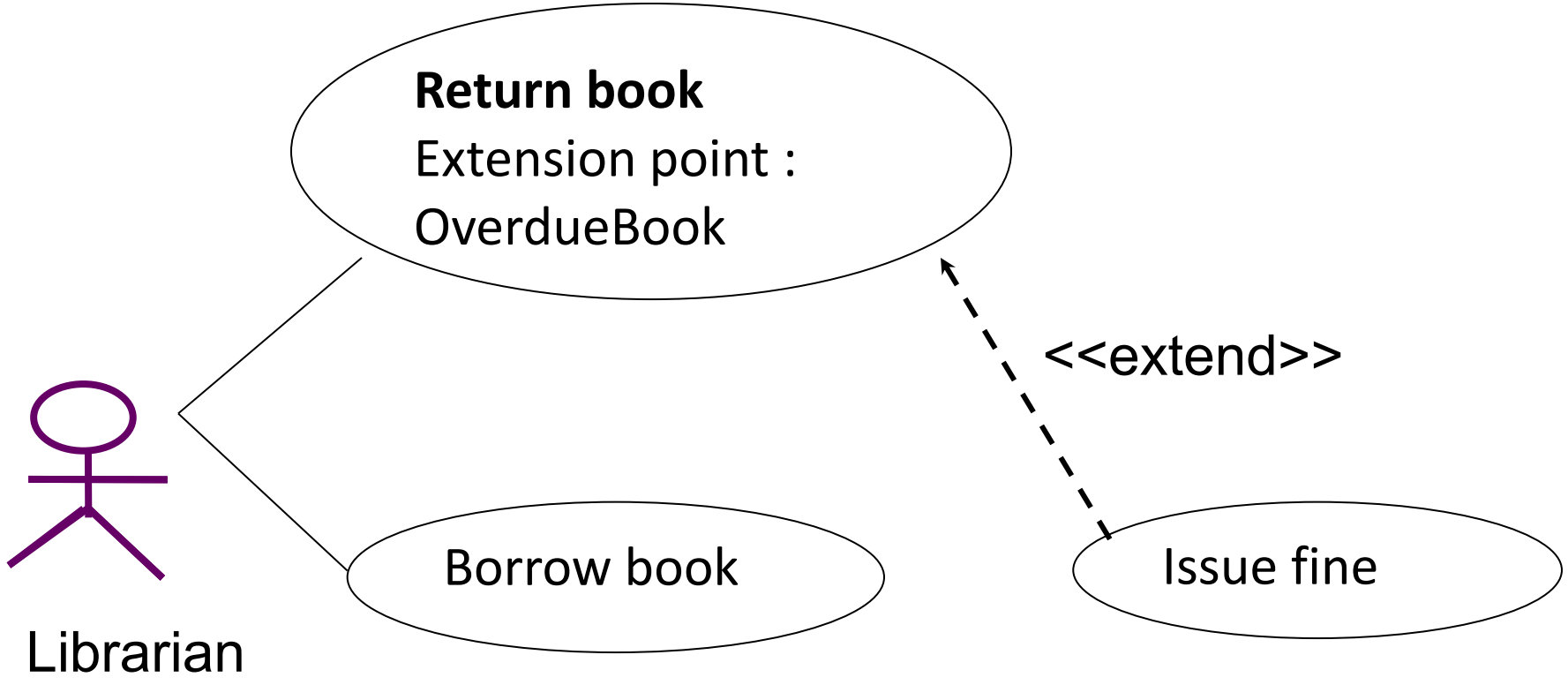
Example of <<extend>> Relationship



<<extend>>

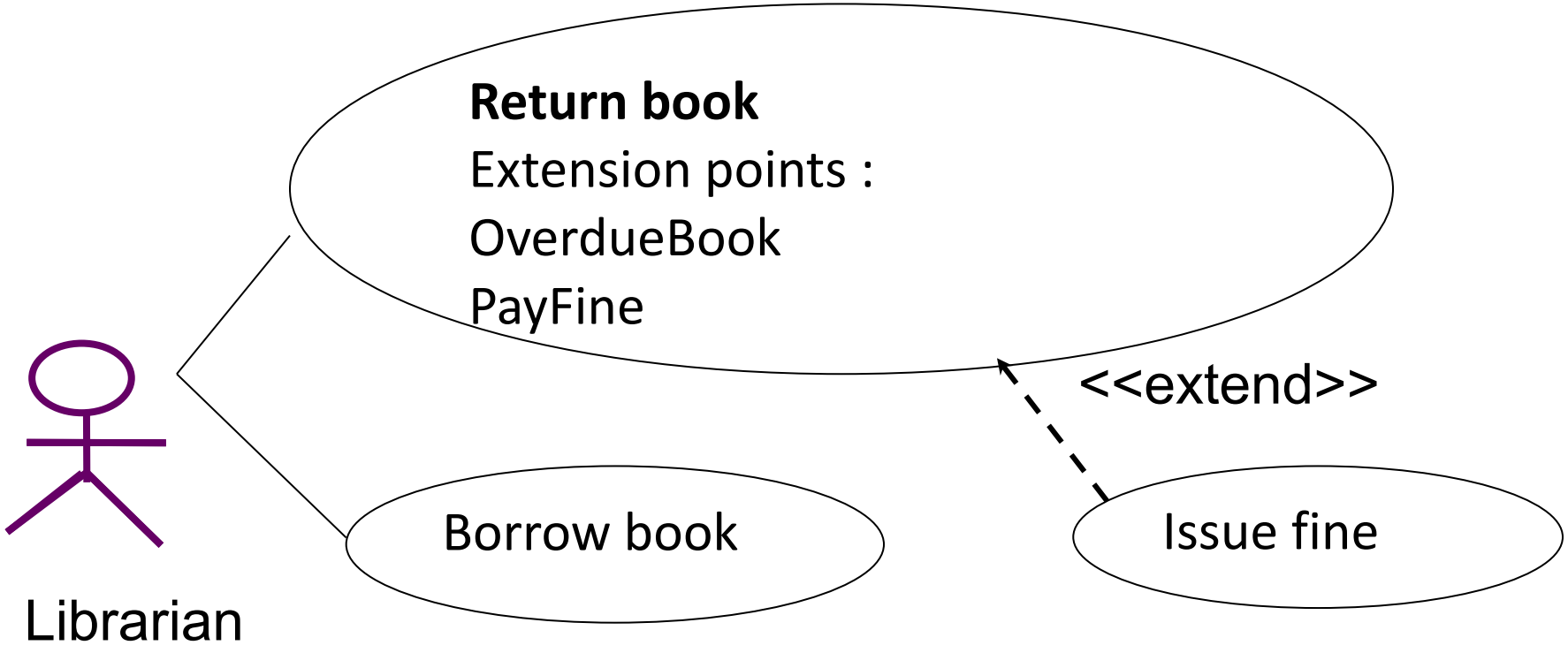
- <<extend>> provides a way to insert new behaviour into an existing use case.
- The base use case provides a set of extension points that are hooks where new behaviour is added.
- Extension use case provides a set of insertion segments that can be inserted into the base use case at these hooks.

Example of **<<extend>>** **Relationship cont..**



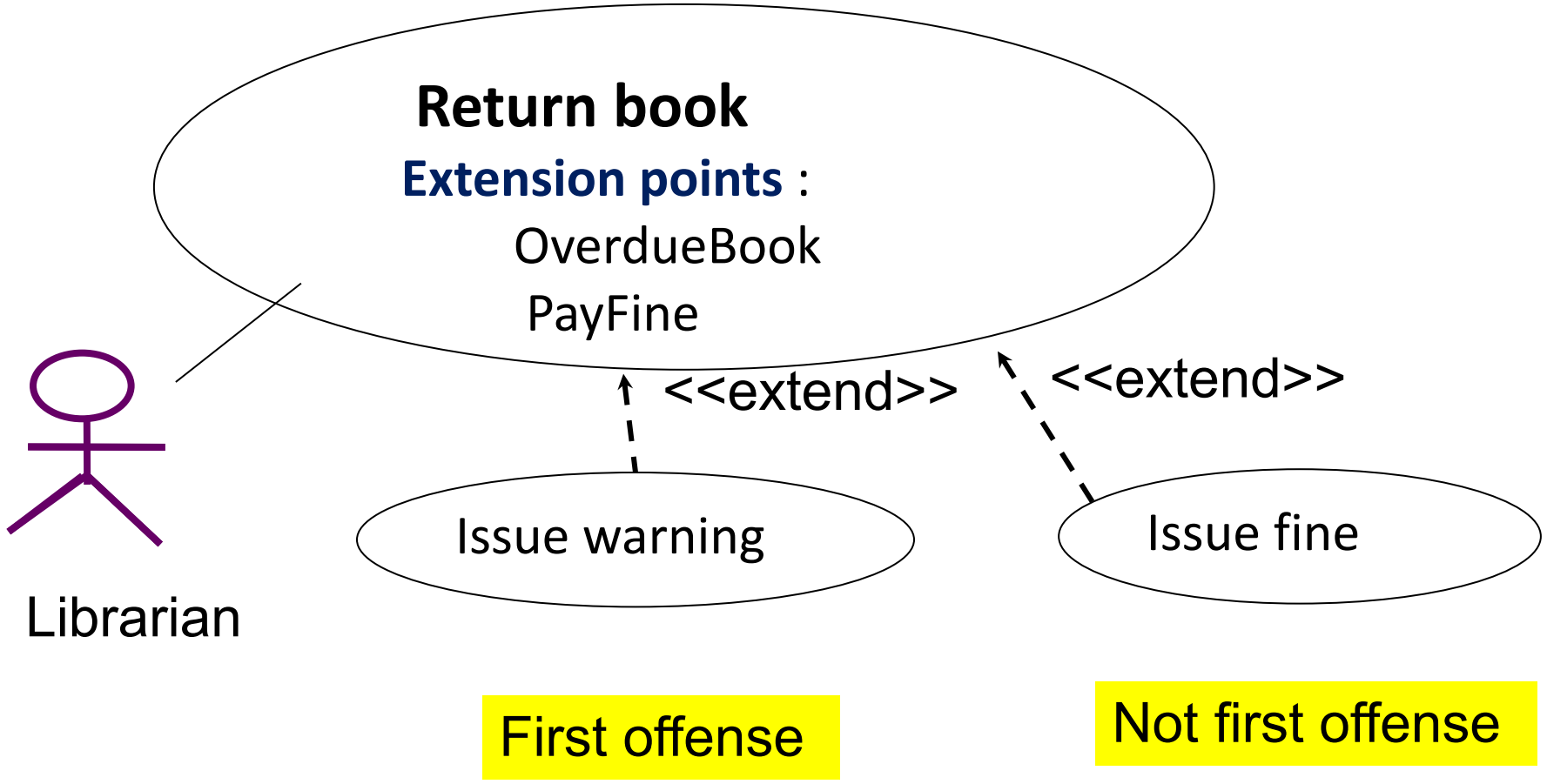
Example of <<extend>> Relationship cont..

- Multiple Insertion segments



Example of <<extend>> Relationship cont..

- Conditional extensions

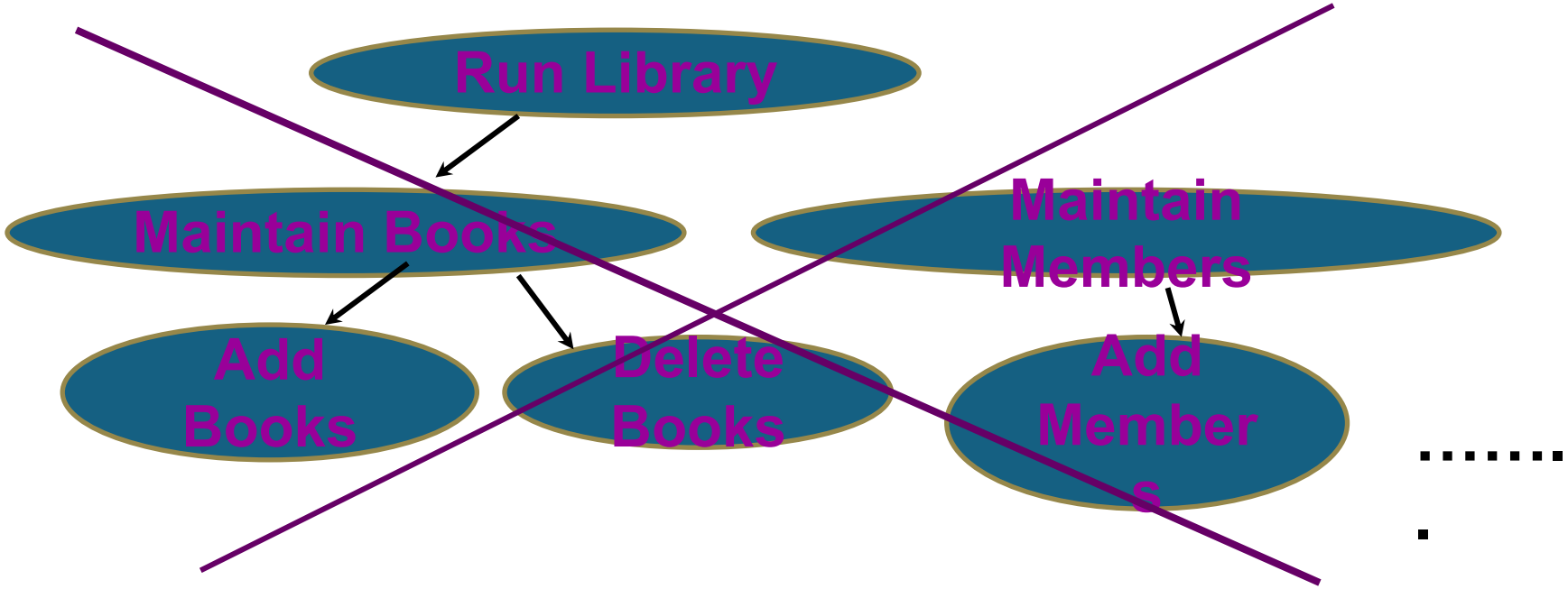


When to apply Use Case Modeling

- When a system is dominated by functional requirements.
- When the system has many type of users to which it delivers different functionality.
- The system has many interfaces.

Hints and Tips for writing Use Cases

- Keep use cases short and simple
- Focus on ***What*** and not *How*.
- Avoid Functional Decomposition



Benefits of Use Case Modelling

- Provides tool for capturing functional requirements.
- Assists in decomposing system into manageable pieces.
- Provides means of communicating with users/stakeholders concerning system functionality in language they understand.
- Provides means of identifying, assigning, tracking, controlling, and managing system development activities.
- Provides aid in estimating project scope, effort, and schedule.

Benefits of Use Case Modelling Cont....

- Aids in defining test plans and test cases.
- Provides baseline for user documentation.
- Provides tool for requirements traceability.
- Provides starting point for identification of data objects or entities.
- Provides specifications for designing user and system interfaces.
- Provides means of defining database access requirements.
- Provides framework for driving the system development project.